

# DERIVATIVES

Calculus is built on two operations — differentiation, which is used to analyse instantaneous rate of change, and integration, which is used to analyse areas. Understanding differentiation and using it to compute derivatives of functions is one of the main aims of this course.

We had a glimpse of derivatives in the previous chapter on limits — in particular Sections 1.1 and 1.2 on tangents and velocities introduced derivatives in disguise. One of the main reasons that we teach limits is to understand derivatives. Fortunately, as we shall see, while one does need to understand limits in order to correctly understand derivatives, one does not need the full machinery of limits in order to *compute and work with* derivatives. The other main part of calculus, integration, we (mostly) leave until a later course.

The derivative finds many applications in many different areas of the sciences. Indeed the reason that calculus is taken by so many university students is so that they may then use the ideas both in subsequent mathematics courses and in other fields. In almost any field in which you study quantitative data you can find calculus lurking somewhere nearby.

Its development came about over a very long time, starting with the ancient Greek geometers. Indian, Persian and Arab mathematicians made significant contributions from around the 6<sup>th</sup> century. But modern calculus really starts with Newton and Leibniz in the 17<sup>th</sup> century who developed independently based on ideas of others including Descartes. Newton applied his work to many physical problems (including orbits of moons and planets) but didn't publish his work. When Leibniz subsequently published his "calculus", Newton accused him of plagiarism — this caused a huge rift between British and continental-European mathematicians which wasn't closed for another century.

## 2.1 ▲ Revisiting Tangent Lines

By way of motivation for the definition of the derivative, we return to the discussion of tangent lines that we started in the previous chapter on limits. We consider, in Exam-

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1 A quick google will turn up many articles on the development and history of calculus. Wikipedia has a good one.

ples [2.1.2](#) and [2.1.3](#), below, the problem of finding the slope of the tangent line to a curve at a point. But let us start by recalling, in Example [2.1.1](#), what is meant by the slope of a straight line.

Example 2.1.1

In this example, we recall what is meant by the slope of the straight line

$$y = \frac{1}{2}x + \frac{3}{2}$$

- We claim that if, as we walk along this straight line, our  $x$ -coordinate changes by an amount  $\Delta x$ , then our  $y$ -coordinate changes by exactly  $\Delta y = \frac{1}{2}\Delta x$ .
- For example, in the figure on the left below, we move from the point

$$(x_0, y_0) = (1, 2 = \frac{1}{2} \times 1 + \frac{3}{2})$$

on the line to the point

$$(x_1, y_1) = (5, 4 = \frac{1}{2} \times 5 + \frac{3}{2})$$

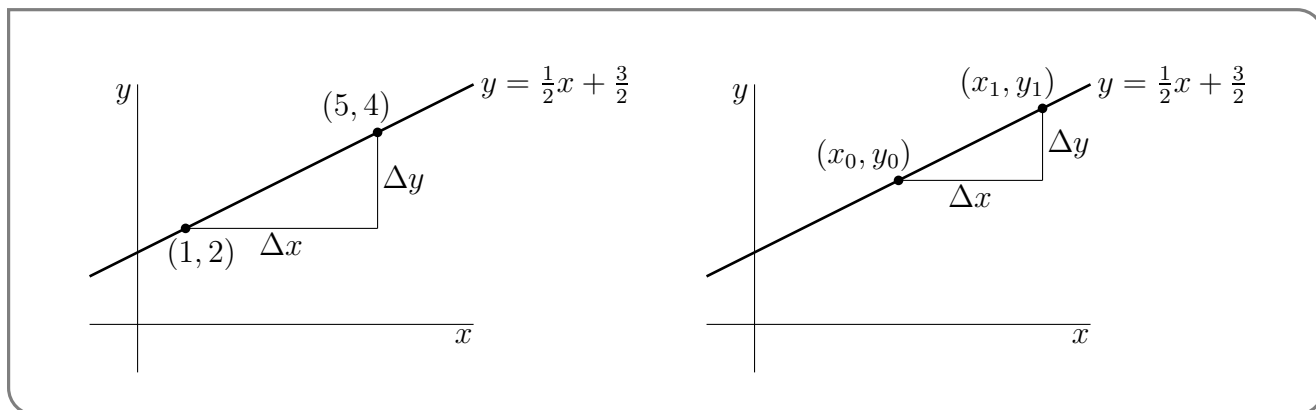
on the line. In this move our  $x$ -coordinate changes by

$$\Delta x = 5 - 1 = 4$$

and our  $y$ -coordinate changes by

$$\Delta y = 4 - 2 = 2$$

which is indeed  $\frac{1}{2} \times 4 = \frac{1}{2}\Delta x$ , as claimed.



- In general, when we move from the point

$$(x_0, y_0) = (x_0, \frac{1}{2}x_0 + \frac{3}{2})$$

on the line to the point

$$(x_1, y_1) = (x_1, \frac{1}{2}x_1 + \frac{3}{2})$$

on the line, our  $x$ -coordinate changes by

$$\Delta x = x_1 - x_0$$

and our  $y$ -coordinate changes by

$$\begin{aligned}\Delta y &= y_1 - y_0 \\ &= \left[\frac{1}{2}x_1 + \frac{3}{2}\right] - \left[\frac{1}{2}x_0 + \frac{3}{2}\right] \\ &= \frac{1}{2}(x_1 - x_0)\end{aligned}$$

which is indeed  $\frac{1}{2}\Delta x$ , as claimed.

- So, for the straight line  $y = \frac{1}{2}x + \frac{3}{2}$ , the ratio  $\frac{\Delta y}{\Delta x} = \frac{y_1 - y_0}{x_1 - x_0}$  always takes the value  $\frac{1}{2}$ , regardless of the choice of initial point  $(x_0, y_0)$  and final point  $(x_1, y_1)$ . This constant ratio is the slope of the line  $y = \frac{1}{2}x + \frac{3}{2}$ .

### Example 2.1.1

Straight lines are special in that for each straight line, there is a fixed number  $m$ , called the slope of the straight line, with the property that if you take *any* two different points,  $(x_0, y_0)$  and  $(x_1, y_1)$ , on the line, the ratio  $\frac{\Delta y}{\Delta x} = \frac{y_1 - y_0}{x_1 - x_0}$ , which is called the rate of change of  $y$  per unit rate of change<sup>2</sup> of  $x$ , always takes the value  $m$ . This is the property that distinguishes lines from other curves.

Other curves do not have this property. In the next two examples we illustrate this point with the parabola  $y = x^2$ . Recall that we studied this example back in Section 1.1. In Example 2.1.2 we find the slope of the tangent line to  $y = x^2$  at a particular point. We generalise this in Example 2.1.3, to show that we can define “the slope of the curve  $y = x^2$ ” at an arbitrary point  $x = x_0$  by considering  $\frac{\Delta y}{\Delta x} = \frac{y_1 - y_0}{x_1 - x_0}$  with  $(x_1, y_1)$  very close to  $(x_0, y_0)$ .

### Example 2.1.2

In this example, let us fix  $(x_0, y_0)$  to be the point  $(2, 4)$  on the parabola  $y = x^2$ . Now let  $(x_1, y_1) = (x_1, x_1^2)$  be some other point on the parabola; that is, a point with  $x_1 \neq x_0$ .

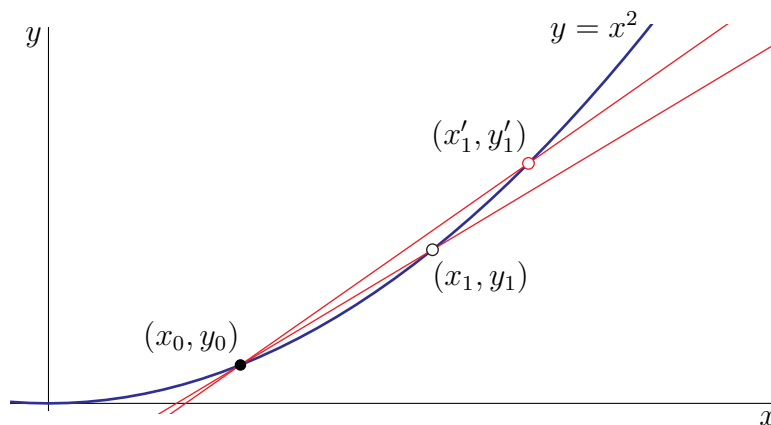
- Draw the straight line through  $(x_0, y_0)$  and  $(x_1, y_1)$  — this is a secant line and we saw these in Chapter 1 when we discussed tangent lines<sup>3</sup>.
- The following table gives the slope,  $\frac{y_1 - y_0}{x_1 - x_0}$ , of the secant line through  $(x_0, y_0) = (2, 4)$  and  $(x_1, y_1)$ , for various different choices of  $(x_1, y_1 = x_1^2)$ .

|                                                         |   |      |      |        |        |   |        |        |      |      |   |
|---------------------------------------------------------|---|------|------|--------|--------|---|--------|--------|------|------|---|
| $x_1$                                                   | 1 | 1.5  | 1.9  | 1.99   | 1.999  | ○ | 2.001  | 2.01   | 2.1  | 2.5  | 3 |
| $y_1 = x_1^2$                                           | 1 | 2.25 | 3.61 | 3.9601 | 3.9960 | ○ | 4.0040 | 4.0401 | 4.41 | 6.25 | 9 |
| $\frac{y_1 - y_0}{x_1 - x_0} = \frac{y_1 - 4}{x_1 - 2}$ | 3 | 3.5  | 3.9  | 3.99   | 3.999  | ○ | 4.001  | 4.01   | 4.1  | 4.5  | 5 |

2 In the “real world” the phrase “rate of change” usually refers to rate of change per unit time. In science it is used more generally.

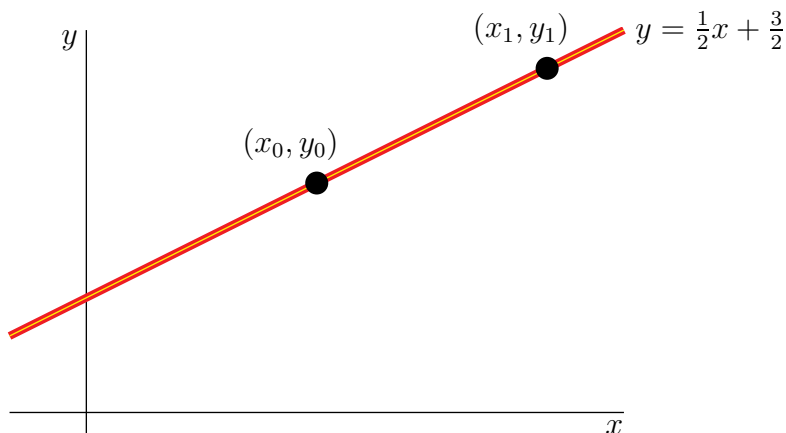
3 If you do not remember this, then please revisit the first couple of sections of Chapter 1.

- So now we have a big table of numbers — what do we do with them? Well, there are messages we can take away from this table.
  - Different choices of  $x_1$  give different values for the slope,  $\frac{y_1 - y_0}{x_1 - x_0}$ , of the secant through  $(x_0, y_0)$  and  $(x_1, y_1)$ . This is illustrated in Figure 2.1.1 below — the slope of the secant through  $(x_0, y_0)$  and  $(x_1, y_1)$  is different from the slope of the secant through  $(x_0, y_0)$  and  $(x'_1, y'_1)$ .

**Figure 2.1.1.**

For a curvy curve, different secants have different slopes.

If the parabola were a straight line this would not be the case — the secant through any two different points on a line is always identical to the line itself and so always has exactly the same slope as the line itself, as is illustrated in Figure 2.1.2 below — the (yellow) secant through  $(x_0, y_0)$  and  $(x_1, y_1)$  lies exactly on top of the (red) line  $y = \frac{1}{2}x + \frac{3}{2}$ .

**Figure 2.1.2.**

For a straight line, all secants have the same slope.

- Now look at the columns of the table closer to the middle. As  $x_1$  gets closer and closer to  $x_0 = 2$ , the slope,  $\frac{y_1 - y_0}{x_1 - x_0}$ , of the secant through  $(x_0, y_0)$  and  $(x_1, y_1)$  appears to get closer and closer to the value 4.

Example 2.1.2

Example 2.1.3

It is very easy to generalise what is happening in Example [2.1.2](#).

- Fix any point  $(x_0, y_0)$  on the parabola  $y = x^2$ . If  $(x_1, y_1)$  is any other point on the parabola  $y = x^2$ , then  $y_1 = x_1^2$  and the slope of the secant through  $(x_0, y_0)$  and  $(x_1, y_1)$  is

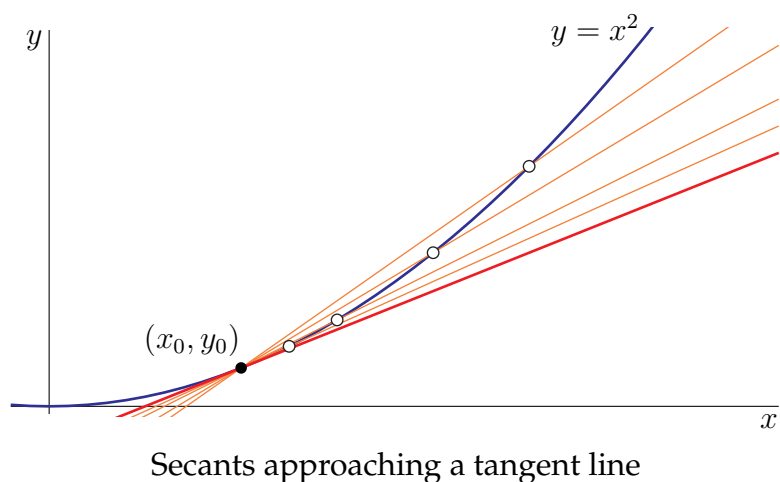
$$\begin{aligned} \text{slope} &= \frac{y_1 - y_0}{x_1 - x_0} = \frac{x_1^2 - x_0^2}{x_1 - x_0} && \text{since } y = x^2 \\ &= \frac{(x_1 - x_0)(x_1 + x_0)}{x_1 - x_0} && \text{remember } a^2 - b^2 = (a - b)(a + b) \\ &= x_1 + x_0 \end{aligned}$$

You should check the values given in the table of Example [2.1.2](#) above to convince yourself that the slope  $\frac{y_1 - y_0}{x_1 - x_0}$  of the secant line really is  $x_0 + x_1 = 2 + x_1$  (since we set  $x_0 = 2$ ).

- Now as we move  $x_1$  closer and closer to  $x_0$ , the slope should move closer and closer to  $2x_0$ . Indeed if we compute the limit carefully — we now have the technology to do this — we see that in the limit as  $x_1 \rightarrow x_0$  the slope becomes  $2x_0$ . That is

$$\begin{aligned} \lim_{x_1 \rightarrow x_0} \frac{y_1 - y_0}{x_1 - x_0} &= \lim_{x_1 \rightarrow x_0} (x_1 + x_0) && \text{by the work we did just above} \\ &= 2x_0 \end{aligned}$$

Taking this limit gives us our first derivative. Of course we haven't yet given the definition of a derivative, so we perhaps wouldn't recognise it yet. We rectify this in the next section.

**Figure 2.1.3.**

- So it is reasonable to say “as  $x_1$  approaches  $x_0$ , the secant through  $(x_0, y_0)$  and  $(x_1, y_1)$  approaches the tangent line to the parabola  $y = x^2$  at  $(x_0, y_0)$ ”. This is what we did back in Section [1.1](#).

The figure above shows four different secants through  $(x_0, y_0)$  for the curve  $y = x^2$ . The four hollow circles are four different choices of  $(x_1, y_1)$ . As  $(x_1, y_1)$  approaches  $(x_0, y_0)$ , the corresponding secant does indeed approach the tangent to  $y = x^2$  at  $(x_0, y_0)$ , which is the heavy (red) straight line in the figure.

Using limits we determined the slope of the tangent line to  $y = x^2$  at  $x_0$  to be  $2x_0$ . Often we will be a little sloppy with our language and instead say “the slope of the parabola  $y = x^2$  at  $(x_0, y_0)$  is  $2x_0$ ” — where we really mean the slope of the line tangent to the parabola at  $x_0$ .

Example 2.1.3

## 2.2 ▲ Definition of the Derivative

We now define the “derivative” explicitly, based on the limiting slope ideas of the previous section. Then we see how to compute some simple derivatives.

Let us now generalise what we did in the last section so as to find “the slope of the curve  $y = f(x)$  at  $(x_0, y_0)$ ” for any smooth enough<sup>4</sup> function  $f(x)$ .

As before, let  $(x_0, y_0)$  be any point on the curve  $y = f(x)$ . So we must have  $y_0 = f(x_0)$ . Now let  $(x_1, y_1)$  be any other point on the same curve. So  $y_1 = f(x_1)$  and  $x_1 \neq x_0$ . Think

<sup>4</sup> The idea of “smooth enough” can be made quite precise. Indeed the word “smooth” has a very precise meaning in mathematics, which we won’t cover here. For now think of “smooth” as meaning roughly just “smooth”.

of  $(x_1, y_1)$  as being pretty close to  $(x_0, y_0)$  so that the difference

$$\Delta x = x_1 - x_0$$

in  $x$ -coordinates is pretty small. In terms of this  $\Delta x$  we have

$$x_1 = x_0 + \Delta x \quad \text{and} \quad y_1 = f(x_0 + \Delta x)$$

We can construct a secant line through  $(x_0, y_0)$  and  $(x_1, y_1)$  just as we did for the parabola above. It has slope

$$\frac{y_1 - y_0}{x_1 - x_0} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

If  $f(x)$  is reasonably smooth<sup>5</sup>, then as  $x_1$  approaches  $x_0$ , i.e. as  $\Delta x$  approaches 0, we would expect the secant through  $(x_0, y_0)$  and  $(x_1, y_1)$  to approach the tangent line to the curve  $y = f(x)$  at  $(x_0, y_0)$ , just as happened in Figure 2.1.3. And more importantly, the slope of the secant through  $(x_0, y_0)$  and  $(x_1, y_1)$  should approach the slope of the tangent line to the curve  $y = f(x)$  at  $(x_0, y_0)$ .

Thus we would expect<sup>6</sup> the slope of the tangent line to the curve  $y = f(x)$  at  $(x_0, y_0)$  to be

$$\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

When we talk of the “slope of the curve” at a point, what we really mean is the slope of the tangent line to the curve at that point. So “the slope of the curve  $y = f(x)$  at  $(x_0, y_0)$ ” is also the limit<sup>7</sup> expressed in the above equation. The derivative of  $f(x)$  at  $x = x_0$  is also defined to be this limit. Which leads<sup>8</sup> us to the most important definition in this text:

5 Again the term “reasonably smooth” can be made more precise.

6 Indeed, we don’t have to expect — it is!

7 This is of course under the assumption that the limit exists — we will talk more about that below.

8 We will rename “ $x_0$ ” to “ $a$ ” and “ $\Delta x$ ” to “ $h$ ”.

**Definition 2.2.1** (Derivative at a point).

Let  $a \in \mathbb{R}$  and let  $f(x)$  be defined on an open interval<sup>9</sup> that contains  $a$ .

- The derivative of  $f(x)$  at  $x = a$  is denoted  $f'(a)$  and is defined by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if the limit exists.

- When the above limit exists, the function  $f(x)$  is said to be differentiable at  $x = a$ . When the limit does not exist, the function  $f(x)$  is said to be not differentiable at  $x = a$ .
- We can equivalently define the derivative  $f'(a)$  by the limit

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

To see that these two definitions are the same, we set  $x = a + h$  and then the limit as  $h$  goes to 0 is equivalent to the limit as  $x$  goes to  $a$ .

Lets now compute the derivatives of some very simple functions. This is our first step towards building up a toolbox for computing derivatives of complicated functions — this process will very much parallel what we did in Chapter 1 with limits. The two simplest functions we know are  $f(x) = c$  and  $g(x) = x$ .

**Example 2.2.2** (Derivative of  $f(x) = c$ )

Let  $a, c \in \mathbb{R}$  be two constants. Compute the derivative of the constant function  $f(x) = c$  at  $x = a$ .

We compute the desired derivative by just substituting the function of interest into the formal definition of the derivative.

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} && \text{(the definition)} \\ &= \lim_{h \rightarrow 0} \frac{c - c}{h} && \text{(substituted in the function)} \\ &= \lim_{h \rightarrow 0} 0 && \text{(simplified things)} \\ &= 0 \end{aligned}$$

**Example 2.2.2**

That was easy! What about the next most complicated function — arguably it's this one:

<sup>9</sup> Recall, from Definition 0.3.5, that the open interval  $(c, d)$  is just the set of all real numbers obeying  $c < x < d$ .

Example 2.2.3 (Derivative of  $g(x) = x$ )

Let  $a \in \mathbb{R}$  and compute the derivative of  $g(x) = x$  at  $x = a$ .

Again, we compute the derivative of  $g$  by just substituting the function of interest into the formal definition of the derivative and then evaluating the resulting limit.

$$\begin{aligned}
 g'(a) &= \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h} && \text{(the definition)} \\
 &= \lim_{h \rightarrow 0} \frac{(a+h) - a}{h} && \text{(substituted in the function)} \\
 &= \lim_{h \rightarrow 0} \frac{h}{h} && \text{(simplified things)} \\
 &= \lim_{h \rightarrow 0} 1 && \text{(simplified a bit more)} \\
 &= 1
 \end{aligned}$$

Example 2.2.3

That was a little harder than the first example, but still quite straight forward — start with the definition and apply what we know about limits.

Thanks to these two examples, we have our first theorem about derivatives:

**Theorem 2.2.4** (Easiest derivatives).

Let  $a, c \in \mathbb{R}$  and let  $f(x) = c$  be the constant function and  $g(x) = x$ . Then

$$f'(a) = 0$$

and

$$g'(a) = 1.$$

To ratchet up the difficulty a little bit more, let us redo the example we have already done a few times  $f(x) = x^2$ . To make it a little more interesting let's change the names of the function and the variable so that it is not exactly the same as Examples [2.1.2](#) and [2.1.3](#).

Example 2.2.5 (Derivative of  $h(t) = t^2$ )

Compute the derivative of

$$h(t) = t^2 \qquad \text{at } t = a$$

- This function isn't quite like the ones we saw earlier — it's a function of  $t$  rather than  $x$ . Recall that a function is a rule which assigns to each input value an output value. So far, we have usually called the input value  $x$ . But this " $x$ " is just a dummy variable representing a generic input value. There is nothing wrong with calling a generic input value  $t$  instead. Indeed, from time to time you will see functions that

are not written as formulas involving  $x$ , but instead are written as formulas in  $t$  (for example representing time — see Section 1.2), or  $z$  (for example representing height), or other symbols.

- So let us write the definition of the derivative

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

and then translate it to the function names and variables at hand:

$$h'(a) = \lim_{h \rightarrow 0} \frac{h(a+h) - h(a)}{h}$$

- But there is a problem — “ $h$ ” plays two roles here — it is both the function name and the small quantity that is going to zero in our limit. It is extremely dangerous to have a symbol represent two different things in a single computation. We need to change one of them. So let’s rename the small quantity that is going to zero in our limit from “ $h$ ” to “ $\Delta t$ ”:

$$h'(a) = \lim_{\Delta t \rightarrow 0} \frac{h(a + \Delta t) - h(a)}{\Delta t}$$

- Now we are ready to begin. Substituting in what the function  $h$  is,

$$\begin{aligned} h'(a) &= \lim_{\Delta t \rightarrow 0} \frac{(a + \Delta t)^2 - a^2}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{a^2 + 2a \Delta t + \Delta t^2 - a^2}{\Delta t} && \text{(just squared out } (a + \Delta t)^2) \\ &= \lim_{\Delta t \rightarrow 0} \frac{2a \Delta t + \Delta t^2}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} (2a + \Delta t) \\ &= 2a \end{aligned}$$

- You should go back check that this is what we got in Example 2.1.3 — just some names have been changed.

Example 2.2.5

### ► An Important Point (and Some Notation)

Notice here that the answer we get depends on our choice of  $a$  — if we want to know the derivative at  $a = 3$  we can just substitute  $a = 3$  into our answer  $2a$  to get the slope is 6. If we want to know at  $a = 1$  (like at the end of Section 1.1) we substitute  $a = 1$  and get the slope is 2. The important thing here is that we can move from the derivative being computed at a specific point to the derivative being a function itself — input any value

of  $a$  and it returns the slope of the tangent line to the curve at the point  $x = a, y = h(a)$ . The variable  $a$  is a dummy variable. We can rename  $a$  to anything we want, like  $x$ , for example. So we can replace every  $a$  in

$$h'(a) = 2a \quad \text{by } x, \text{ giving} \quad h'(x) = 2x$$

where all we have done is replaced the symbol  $a$  by the symbol  $x$ .

We can do this more generally and tweak the derivative at a specific point  $a$  to obtain the derivative as a function of  $x$ . We replace

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

with

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

which gives us the following definition

**Definition 2.2.6** (Derivative as a function).

Let  $f(x)$  be a function.

- The derivative of  $f(x)$  with respect to  $x$  is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists.

- If the derivative  $f'(x)$  exists for all  $x \in (a, b)$  we say that  $f$  is differentiable on  $(a, b)$ .
- Note that we will sometimes be a little sloppy with our discussions and simply write “ $f$  is differentiable” to mean “ $f$  is differentiable on an interval we are interested in” or “ $f$  is differentiable everywhere”.

Notice that we are no longer thinking of tangent lines, rather this is an operation we can do on a function. For example:

**Example 2.2.7** (The derivative of  $f(x) = \frac{1}{x}$ )

Let  $f(x) = \frac{1}{x}$  and compute its derivative with respect to  $x$  — think carefully about where the derivative exists.

- Our first step is to write down the definition of the derivative — at this stage, we know of no other strategy for computing derivatives.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{(the definition)}$$

- And now we substitute in the function and compute the limit.

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} && \text{(the definition)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{1}{x+h} - \frac{1}{x} \right] && \text{(substituted in the function)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \frac{x - (x+h)}{x(x+h)} && \text{(wrote over a common denominator)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \frac{-h}{x(x+h)} && \text{(started cleanup)} \\
 &= \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} \\
 &= -\frac{1}{x^2}
 \end{aligned}$$

- Notice that the original function  $f(x) = \frac{1}{x}$  was not defined at  $x = 0$  and the derivative is also not defined at  $x = 0$ . This does happen more generally — if  $f(x)$  is not defined at a particular point  $x = a$ , then the derivative will not exist at that point either.

Example 2.2.7

So we now have two slightly different ideas of derivatives:

- The derivative  $f'(a)$  at a specific point  $x = a$ , being the slope of the tangent line to the curve at  $x = a$ , and
- The derivative as a function,  $f'(x)$  as defined in Definition [2.2.6](#).

Of course, if we have  $f'(x)$  then we can always recover the derivative at a specific point by substituting  $x = a$ .

As we noted at the beginning of the chapter, the derivative was discovered independently by Newton and Leibniz in the late 17<sup>th</sup> century. Because their discoveries were independent, Newton and Leibniz did not have exactly the same notation. Stemming from this, and from the many different contexts in which derivatives are used, there are quite a few alternate notations for the derivative:

**Notation 2.2.8.**

The following notations are all used for “the derivative of  $f(x)$  with respect to  $x$ ”

$$f'(x) \quad \frac{df}{dx} \quad \frac{d}{dx}f(x) \quad \dot{f}(x) \quad Df(x) \quad D_x f(x),$$

while the following notations are all used for “the derivative of  $f(x)$  at  $x = a$ ”

$$f'(a) \quad \frac{df}{dx}(a) \quad \frac{d}{dx}f(x) \Big|_{x=a} \quad \dot{f}(a) \quad Df(a) \quad D_x f(a).$$

Some things to note about these notations:

- We will generally use the first three, but you should recognise them all. The notation  $f'(a)$  is due to Lagrange, while the notation  $\frac{df}{dx}(a)$  is due to Leibniz. They are both very useful. Neither can be considered “better”.
- Leibniz notation writes the derivative as a “fraction” — however it is definitely not a fraction and should not be thought of in that way. It is just shorthand, which is read as “the derivative of  $f$  with respect to  $x$ ”.
- You read  $f'(x)$  as “ $f$ -prime of  $x$ ”, and  $\frac{df}{dx}$  as “dee- $f$ -dee- $x$ ”, and  $\frac{d}{dx}f(x)$  as “dee-by-dee- $x$  of  $f$ ”.
- Similarly you read  $\frac{df}{dx}(a)$  as “dee- $f$ -dee- $x$  at  $a$ ”, and  $\frac{d}{dx}f(x) \Big|_{x=a}$  as “dee-by-dee  $x$  of  $f$  at  $x$  equals  $a$ ”.
- The notation  $\dot{f}$  is due to Newton. In physics, it is common to use  $\dot{f}(t)$  to denote the derivative of  $f$  with respect to time.

## » Back to Computing Some Derivatives

At this point we could try to start working out how derivatives interact with arithmetic and make an “Arithmetic of derivatives” theorem just like the one we saw for limits (Theorem 2). We will get there shortly, but before that it is important that we become more comfortable with computing derivatives using limits and then understanding what the derivative actually means. So — more examples.

### Example 2.2.9 $\left(\frac{d}{dx}\sqrt{x}\right)$

Compute the derivative,  $f'(a)$ , of the function  $f(x) = \sqrt{x}$  at the point  $x = a$  for any  $a > 0$ .

- So again we start with the definition of derivative and go from there:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{\sqrt{x} - \sqrt{a}}{x - a}$$

- As  $x$  tends to  $a$ , the numerator and denominator both tend to zero. But  $\frac{0}{0}$  is not defined. So to get a well defined limit we need to exhibit a cancellation between the numerator and denominator — just as we saw in Examples 1.4.11 and 1.4.16. Now there are two equivalent ways to proceed from here, both based on a similar “trick”.
- For the first, review Example 1.4.16, which concerned taking a limit involving square-roots, and recall that we used “multiplication by the conjugate” there:

$$\begin{aligned}\frac{\sqrt{x} - \sqrt{a}}{x - a} &= \frac{\sqrt{x} - \sqrt{a}}{x - a} \times \frac{\sqrt{x} + \sqrt{a}}{\sqrt{x} + \sqrt{a}} \quad \left( \text{multiplication by } 1 = \frac{\text{conjugate}}{\text{conjugate}} \right) \\ &= \frac{(\sqrt{x} - \sqrt{a})(\sqrt{x} + \sqrt{a})}{(x - a)(\sqrt{x} + \sqrt{a})} \\ &= \frac{x - a}{(x - a)(\sqrt{x} + \sqrt{a})} \quad (\text{since } (A - B)(A + B) = A^2 - B^2) \\ &= \frac{1}{\sqrt{x} + \sqrt{a}}\end{aligned}$$

- Alternatively, we can arrive at  $\frac{\sqrt{x} - \sqrt{a}}{x - a} = \frac{1}{\sqrt{x} + \sqrt{a}}$  by using almost the same trick to factor the denominator. Just set  $A = \sqrt{x}$  and  $B = \sqrt{a}$  in  $A^2 - B^2 = (A - B)(A + B)$  to get

$$x - a = (\sqrt{x} - \sqrt{a})(\sqrt{x} + \sqrt{a})$$

and then substitute this little fact into our expression

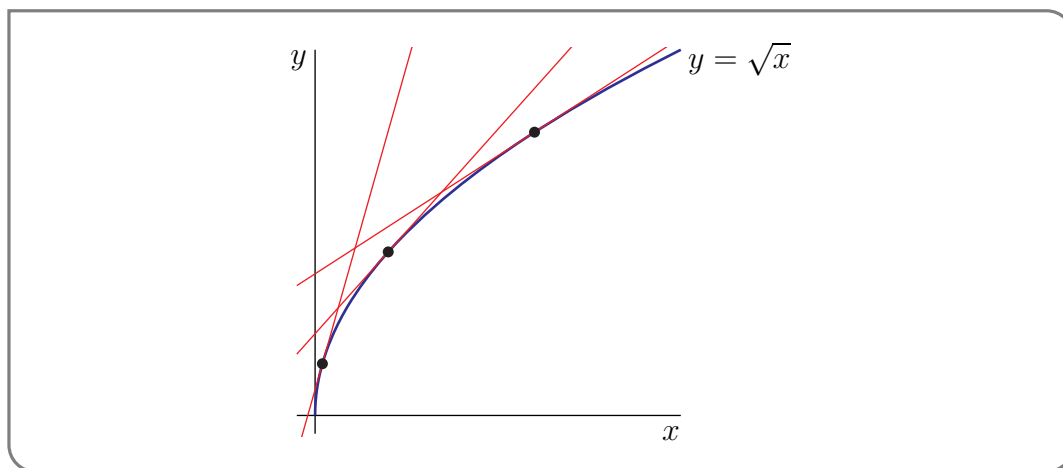
$$\begin{aligned}\frac{\sqrt{x} - \sqrt{a}}{x - a} &= \frac{\sqrt{x} - \sqrt{a}}{(\sqrt{x} - \sqrt{a})(\sqrt{x} + \sqrt{a})} \quad (\text{now cancel common factors}) \\ &= \frac{1}{(\sqrt{x} + \sqrt{a})}\end{aligned}$$

- Once we know that  $\frac{\sqrt{x} - \sqrt{a}}{x - a} = \frac{1}{\sqrt{x} + \sqrt{a}}$ , we can take the limit we need:

$$\begin{aligned}f'(a) &= \lim_{x \rightarrow a} \frac{\sqrt{x} - \sqrt{a}}{x - a} \\ &= \lim_{x \rightarrow a} \frac{1}{\sqrt{x} + \sqrt{a}} \\ &= \frac{1}{2\sqrt{a}}\end{aligned}$$

- We should think about the domain of  $f'$  here — that is, for which values of  $a$  is  $f'(a)$  defined? The original function  $f(x)$  was defined for all  $x \geq 0$ , however the derivative  $f'(a) = \frac{1}{2\sqrt{a}}$  is undefined at  $a = 0$ .

If we draw a careful picture of  $\sqrt{x}$  around  $x = 0$  we can see why this has to be the case. The figure below shows three different tangent lines to the graph of  $y = f(x) = \sqrt{x}$ . As the point of tangency moves closer and closer to the origin, the tangent line gets steeper and steeper. The slope of the tangent line at  $(a, \sqrt{a})$  blows up as  $a \rightarrow 0$ .



Example 2.2.9

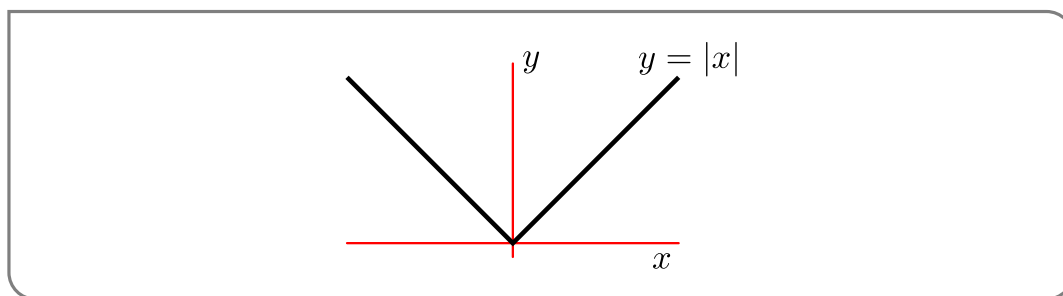
Example 2.2.10  $\left(\frac{d}{dx} \{|x|\}\right)$ 

Compute the derivative,  $f'(a)$ , of the function  $f(x) = |x|$  at the point  $x = a$ .

- We should start this example by recalling the definition of  $|x|$  (we saw this back in Example [1.5.6](#)):

$$|x| = \begin{cases} -x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ x & \text{if } x > 0. \end{cases}$$

It is definitely not just “chop off the minus sign”.



- This breaks our computation of the derivative into 3 cases depending on whether  $x$  is positive, negative or zero.
- Assume  $x > 0$ . Then

$$\begin{aligned} \frac{df}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} \end{aligned}$$

Since  $x > 0$  and we are interested in the behaviour of this function as  $h \rightarrow 0$  we can assume  $h$  is much smaller than  $x$ . This means  $x + h > 0$  and so  $|x + h| = x + h$ .

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{x + h - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = 1 \end{aligned} \quad \text{as expected}$$

- Assume  $x < 0$ . Then

$$\begin{aligned} \frac{df}{dx} &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{|x + h| - |x|}{h} \end{aligned}$$

Since  $x < 0$  and we are interested in the behaviour of this function as  $h \rightarrow 0$  we can assume  $h$  is much smaller than  $x$ . This means  $x + h < 0$  and so  $|x + h| = -(x + h)$ .

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{-(x + h) - (-x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h} = -1 \end{aligned}$$

- When  $x = 0$  we have

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{|0 + h| - |0|}{h} \\ &= \lim_{h \rightarrow 0} \frac{|h|}{h} \end{aligned}$$

To proceed we need to know if  $h > 0$  or  $h < 0$ , so we must use one-sided limits. The limit from above is:

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{|h|}{h} &= \lim_{h \rightarrow 0^+} \frac{h}{h} && \text{since } h > 0, |h| = h \\ &= 1 \end{aligned}$$

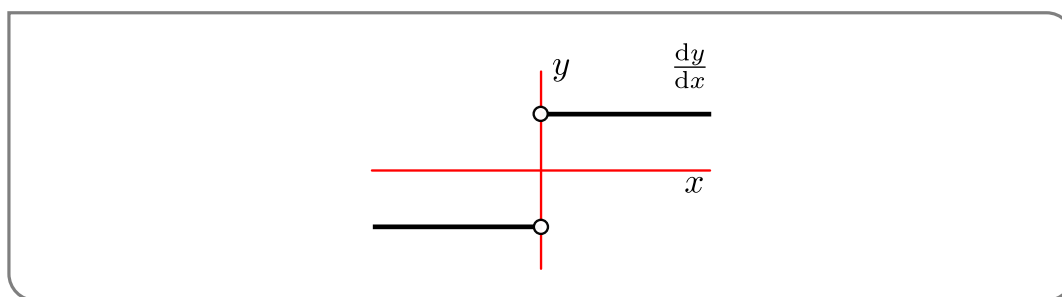
Whereas, the limit from below is:

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{|h|}{h} &= \lim_{h \rightarrow 0^-} \frac{-h}{h} && \text{since } h < 0, |h| = -h \\ &= -1 \end{aligned}$$

Since the one-sided limits differ, the limit as  $h \rightarrow 0$  does not exist. And thus the derivative does not exist as  $x = 0$ .

In summary:

$$\frac{d}{dx}|x| = \begin{cases} -1 & \text{if } x < 0 \\ DNE & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$$



Example 2.2.10

### ► Where is the Derivative Undefined?

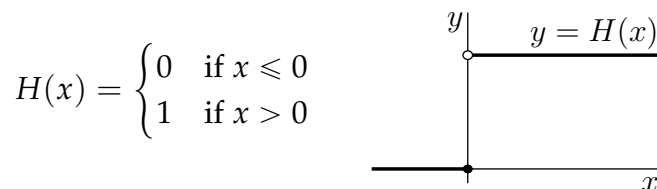
According to Definition 2.2.1, the derivative  $f'(a)$  exists precisely when the limit  $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$  exists. That limit is also the slope of the tangent line to the curve  $y = f(x)$  at  $x = a$ . That limit does not exist when the curve  $y = f(x)$  does not have a tangent line at  $x = a$  or when the curve does have a tangent line, but the tangent line has infinite slope. We have already seen some examples of this.

- In Example 2.2.7, we considered the function  $f(x) = \frac{1}{x}$ . This function “blows up” (i.e. becomes infinite) at  $x = 0$ . It does not have a tangent line at  $x = 0$  and its derivative does not exist at  $x = 0$ .
- In Example 2.2.10, we considered the function  $f(x) = |x|$ . This function does not have a tangent line at  $x = 0$ , because there is a sharp corner in the graph of  $y = |x|$  at  $x = 0$ . (Look at the graph in Example 2.2.10.) So the derivative of  $f(x) = |x|$  does not exist at  $x = 0$ .

Here are a few more examples.

Example 2.2.11

Visually, the function



does not have a tangent line at  $(0,0)$ . Not surprisingly, when  $a = 0$  and  $h$  tends to 0 with  $h > 0$ ,

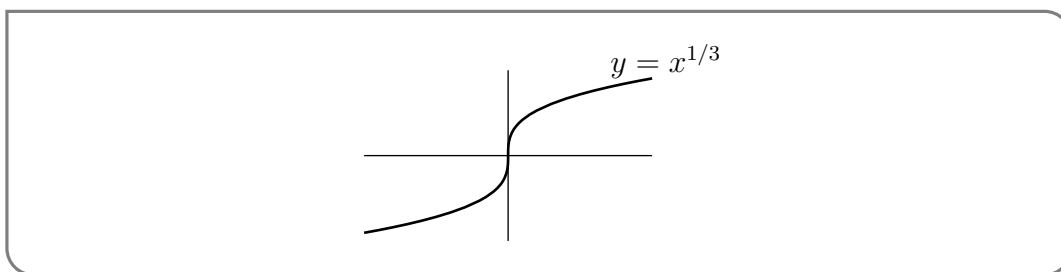
$$\frac{H(a+h) - H(a)}{h} = \frac{H(h) - H(0)}{h} = \frac{1}{h}$$

blows up. The same sort of computation shows that  $f'(a)$  cannot possibly exist whenever the function  $f$  is not continuous at  $a$ . We will formalize, and prove, this statement in Theorem 2.2.14, below.

## Example 2.2.11

Example 2.2.12  $\left(\frac{d}{dx}x^{1/3}\right)$ 

Visually, it looks like the function  $f(x) = x^{1/3}$ , sketched below, (this might be a good point to recall that cube roots of negative numbers are negative — for example, since  $(-1)^3 = -1$ , the cube root of  $-1$  is  $-1$ ),



has the  $y$ -axis as its tangent line at  $(0,0)$ . So we would expect that  $f'(0)$  does not exist. Let's check. With  $a = 0$ ,

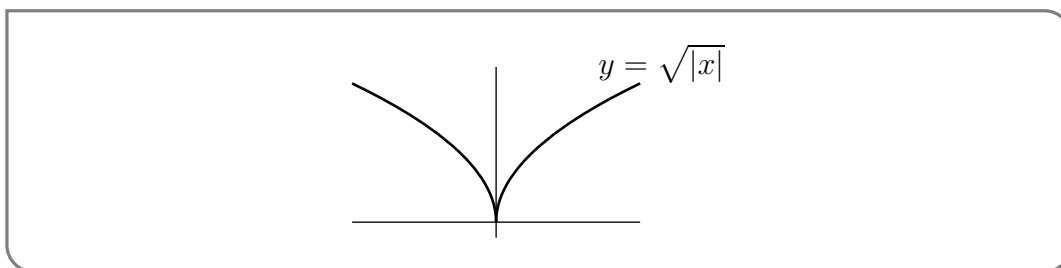
$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^{1/3}}{h} = \lim_{h \rightarrow 0} \frac{1}{h^{2/3}} = DNE$$

as expected.

## Example 2.2.12

Example 2.2.13  $\left(\frac{d}{dx}\sqrt{|x|}\right)$ 

We have already considered the derivative of the function  $\sqrt{x}$  in Example 2.2.9. We'll now look at the function  $f(x) = \sqrt{|x|}$ . Recall, from Example 2.2.10, the definition of  $|x|$ . When  $x > 0$ , we have  $|x| = x$  and  $f(x)$  is identical to  $\sqrt{x}$ . When  $x < 0$ , we have  $|x| = -x$  and  $f(x) = \sqrt{-x}$ . So to graph  $y = \sqrt{|x|}$  when  $x < 0$ , you just have to graph  $y = \sqrt{x}$  for  $x > 0$  and then send  $x \rightarrow -x$  — i.e. reflect the graph in the  $y$ -axis. Here is the graph. The pointy



thing at the origin is called a cusp. The graph of  $y = f(x)$  does not have a tangent line at  $(0,0)$  and, correspondingly,  $f'(0)$  does not exist because

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{\sqrt{|h|}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{\sqrt{h}} = DNE$$

Example 2.2.13

**Theorem 2.2.14.**

If the function  $f(x)$  is differentiable at  $x = a$ , then  $f(x)$  is also continuous at  $x = a$ .

*Proof.* The function  $f(x)$  is continuous at  $x = a$  if and only if the limit of

$$f(a+h) - f(a) = \frac{f(a+h) - f(a)}{h} h$$

as  $h \rightarrow 0$  exists and is zero. But if  $f(x)$  is differentiable at  $x = a$ , then, as  $h \rightarrow 0$ , the first factor,  $\frac{f(a+h)-f(a)}{h}$  converges to  $f'(a)$  and the second factor,  $h$ , converges to zero. So the product provision of our arithmetic of limits Theorem 1.4.2 implies that the product  $\frac{f(a+h)-f(a)}{h} h$  converges to  $f'(a) \cdot 0 = 0$  too.  $\square$

Notice that while this theorem is useful as stated, it is (arguably) more often applied in its contrapositive<sup>10</sup> form:

**Theorem 2.2.15** (The contrapositive of Theorem 2.2.14).

If  $f(x)$  is not continuous at  $x = a$  then it is not differentiable at  $x = a$ .

As the above examples illustrate, this statement does not tell us what happens if  $f$  is continuous at  $x = a$  — we have to think!

## 2.3 ▲ Interpretations of the Derivative

In the previous sections we defined the derivative as the slope of a tangent line, using a particular limit. This allows us to compute “the slope of a curve<sup>11</sup>” and provides us with one interpretation of the derivative. However, the main importance of derivatives does not come from this application. Instead, (arguably) it comes from the interpretation of the derivative as the instantaneous rate of change of a quantity.

10 If you have forgotten what the contrapositive is, then quickly reread Footnote<sup>10</sup> in Section 1.3.

11 Again — recall that we are being a little sloppy with this term — we really mean “The slope of the tangent line to the curve”.

### ► Instantaneous Rate of Change

In fact we have already (secretly) used a derivative to compute an instantaneous rate of change in Section 1.2. For your convenience we'll review that computation here, in Example 2.3.1, and then generalise it.

#### Example 2.3.1

You drop a ball from a tall building. After  $t$  seconds the ball has fallen a distance of  $s(t) = 4.9t^2$  metres. What is the velocity of the ball one second after it is dropped?

- In the time interval from  $t = 1$  to  $t = 1 + h$  the ball travels a distance

$$s(1 + h) - s(1) = 4.9(1 + h)^2 - 4.9(1)^2 = 4.9[2h + h^2]$$

- So the average velocity over this time interval is

$$\begin{aligned} & \text{average velocity from } t = 1 \text{ to } t = 1 + h \\ &= \frac{\text{distance travelled from } t = 1 \text{ to } t = 1 + h}{\text{length of time from } t = 1 \text{ to } t = 1 + h} \\ &= \frac{s(1 + h) - s(1)}{h} \\ &= \frac{4.9[2h + h^2]}{h} \\ &= 4.9[2 + h] \end{aligned}$$

- The instantaneous velocity at time  $t = 1$  is then defined to be the limit

$$\begin{aligned} & \text{instantaneous velocity at time } t = 1 \\ &= \lim_{h \rightarrow 0} [\text{average velocity from } t = 1 \text{ to } t = 1 + h] \\ &= \lim_{h \rightarrow 0} \frac{s(1 + h) - s(1)}{h} = s'(1) \\ &= \lim_{h \rightarrow 0} 4.9[2 + h] \\ &= 9.8\text{m/sec} \end{aligned}$$

- We conclude that the instantaneous velocity at time  $t = 1$ , which is the instantaneous rate of change of distance per unit time at time  $t = 1$ , is the derivative  $s'(1) = 9.8\text{m/sec}$ .

#### Example 2.3.1

Now suppose, more generally, that you are taking a walk and that as you walk, you are continuously measuring some quantity, like temperature, and that the measurement

at time  $t$  is  $f(t)$ . Then the

$$\begin{aligned} & \text{average rate of change of } f(t) \text{ from } t = a \text{ to } t = a + h \\ &= \frac{\text{change in } f(t) \text{ from } t = a \text{ to } t = a + h}{\text{length of time from } t = a \text{ to } t = a + h} \\ &= \frac{f(a + h) - f(a)}{h} \end{aligned}$$

so the

$$\begin{aligned} & \text{instantaneous rate of change of } f(t) \text{ at } t = a \\ &= \lim_{h \rightarrow 0} [\text{average rate of change of } f(t) \text{ from } t = a \text{ to } t = a + h] \\ &= \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \\ &= f'(a) \end{aligned}$$

In particular, if you are walking along the  $x$ -axis and your  $x$ -coordinate at time  $t$  is  $x(t)$ , then  $x'(a)$  is the instantaneous rate of change (per unit time) of your  $x$ -coordinate at time  $t = a$ , which is your velocity at time  $a$ . If  $v(t)$  is your velocity at time  $t$ , then  $v'(a)$  is the instantaneous rate of change of your velocity at time  $a$ . This is called your acceleration at time  $a$ .

You might expect that if the instantaneous rate of change of a function at time  $c$  is strictly positive, then, in some sense, the function is increasing at  $t = c$ . You would be right. Indeed, if  $f'(c) > 0$ , then, by definition, the limit of  $\frac{f(t) - f(c)}{t - c}$  as  $t$  approaches  $c$  is strictly bigger than zero. So

- for all  $t > c$  that are sufficiently close<sup>12</sup> to  $c$

$$\begin{aligned} \frac{f(t) - f(c)}{t - c} > 0 &\implies f(t) - f(c) > 0 && (\text{since } t - c > 0) \\ &\implies f(t) > f(c) \end{aligned}$$

- for all  $t < c$  that are sufficiently close to  $c$

$$\begin{aligned} \frac{f(t) - f(c)}{t - c} > 0 &\implies f(t) - f(c) < 0 && (\text{since } t - c < 0) \\ &\implies f(t) < f(c) \end{aligned}$$

Consequently we say that “ $f(t)$  is increasing at  $t = c$ ”. If we wish to emphasise that the inequalities above are the strict inequalities  $>$  and  $<$ , as opposed to  $\geq$  and  $\leq$ , we will say that “ $f(t)$  is strictly increasing at  $t = c$ ”.

12 This is typical mathematician speak — it allows us to be completely correct, without being terribly precise. In this context, “sufficiently close” means “The following need not be true for all  $t$  bigger than  $c$ , but there must exist some  $b > c$  so that the following is true for all  $c < t < b$ ”. Typically we do not know what  $b$  is. And typically it does not matter what the exact value of  $b$  is. All that matters is that  $b$  exists and is strictly bigger than  $c$ .

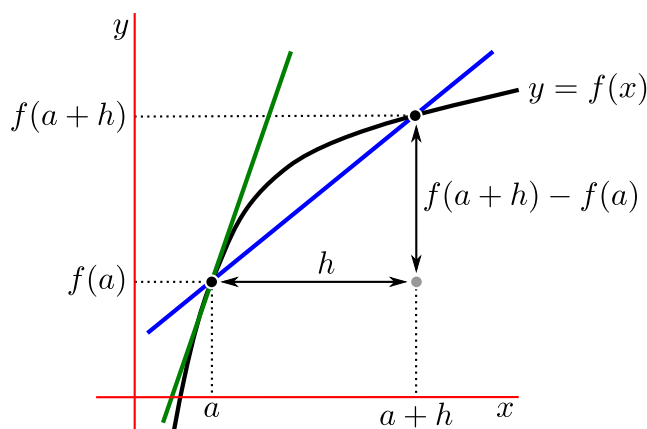
## ► Slope

Suppose that  $y = f(x)$  is the equation of a curve in the  $xy$ -plane. That is,  $f(x)$  is the  $y$ -coordinate of the point on the curve whose  $x$ -coordinate is  $x$ . Then, as we have already seen,

$$[\text{the slope of the secant through } (a, f(a)) \text{ and } (a+h, f(a+h))] = \frac{f(a+h) - f(a)}{h}$$

This is shown in Figure 2.3.1 below.

**Figure 2.3.1.**



In order to create the tangent line (as we have done a few times now) we squeeze  $h \rightarrow 0$ . As we do this, the secant through  $(a, f(a))$  and  $(a+h, f(a+h))$  approaches<sup>13</sup> the tangent line to  $y = f(x)$  at  $x = a$ . Since the secant becomes the tangent line in this limit, the slope of the secant becomes the slope of the tangent and

$$[\text{the slope of the tangent line to } y = f(x) \text{ at } x = a] = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f'(a).$$

Let us go a little further and work out a general formula for the equation of the tangent line to  $y = f(x)$  at  $x = a$ . We know that the tangent line

- has slope  $f'(a)$  and
- passes through the point  $(a, f(a))$ .

There are a couple of different ways to construct the equation of the tangent line from this information. One is to observe, as in Figure 2.3.2 that if  $(x, y)$  is any other point on the tangent line then the line segment from  $(a, f(a))$  to  $(x, y)$  is part of the tangent line and so also has slope  $f'(a)$ . That is,

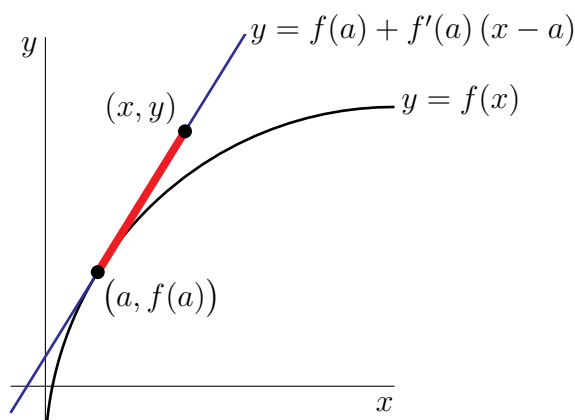
$$\frac{y - f(a)}{x - a} = [\text{the slope of the tangent line}] = f'(a)$$

<sup>13</sup> We are of course assuming that the curve is smooth enough to have a tangent line at  $a$ .

Cross multiplying gives us the equation of the tangent line:

$$y - f(a) = f'(a)(x - a) \quad \text{or} \quad y = f(a) + f'(a)(x - a)$$

**Figure 2.3.2.**



A line segment of a tangent line

A second way to derive the same equation of the same tangent line is to recall that the general equation for a line, with finite slope, is  $y = mx + b$ , where  $m$  is the slope and  $b$  is the  $y$ -intercept. We already know the slope — so  $m = f'(a)$ . To work out  $b$  we use the other piece of information —  $(a, f(a))$  is on the line. So  $(x, y) = (a, f(a))$  must solve  $y = f'(a)x + b$ . That is,

$$f(a) = f'(a) \cdot a + b \quad \text{and so} \quad b = f(a) - af'(a)$$

Hence our equation is, once again,

$$\begin{aligned} y &= f'(a) \cdot x + (f(a) - af'(a)) && \text{or, after rearranging a little,} \\ y &= f(a) + f'(a)(x - a) \end{aligned}$$

This is a very useful formula, so perhaps we should make it a theorem.

**Theorem 2.3.2 (Tangent line).**

The tangent line to the curve  $y = f(x)$  at  $x = a$  is given by the equation

$$y = f(a) + f'(a)(x - a)$$

provided the derivative  $f'(a)$  exists.

# DERIVATIVES

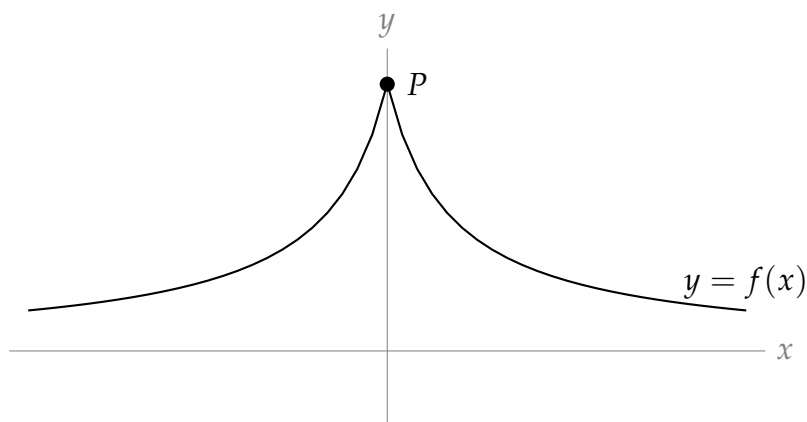
## 2.1▲ Revisiting tangent lines

### Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

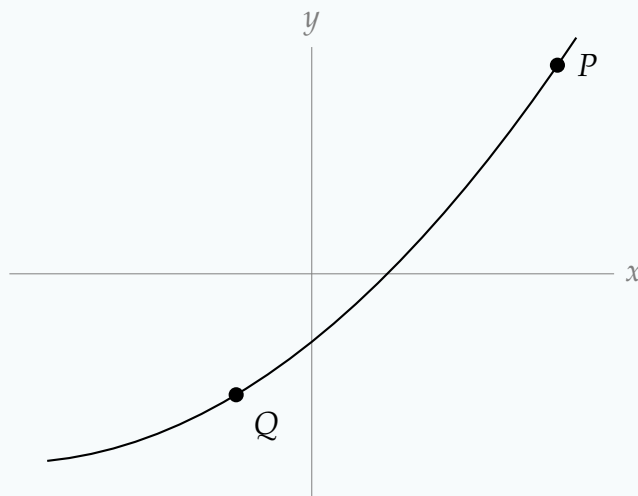
### ►► Stage 1

Q[1]: Shown below is the graph  $y = f(x)$ . If we choose a point  $Q$  on the graph to the *left* of the  $y$ -axis, is the slope of the secant line through  $P$  and  $Q$  positive or negative? If we choose a point  $Q$  on the graph to the *right* of the  $y$ -axis, is the slope of the secant line through  $P$  and  $Q$  positive or negative?

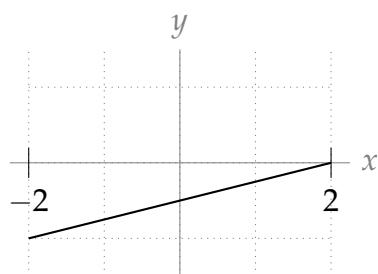


Q[2]: Shown below is the graph  $y = f(x)$ .

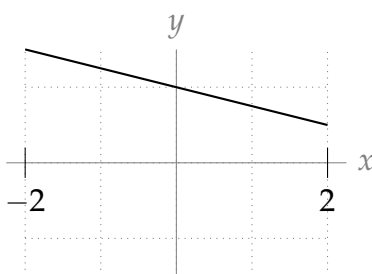
- (a) If we want the slope of the secant line through  $P$  and  $Q$  to *increase*, should we slide  $Q$  closer to  $P$ , or further away?
- (b) Which is larger, the slope of the tangent line at  $P$ , or the slope of the secant line through  $P$  and  $Q$ ?



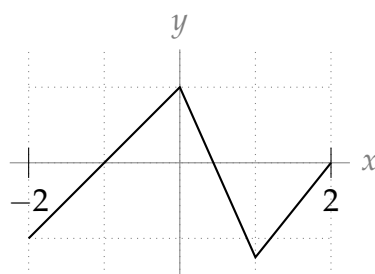
Q[3]: Group the functions below into collections whose secant lines from  $x = -2$  to  $x = 2$  all have the same slopes.



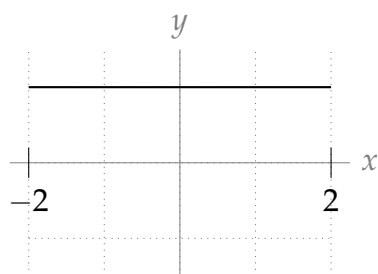
(a)



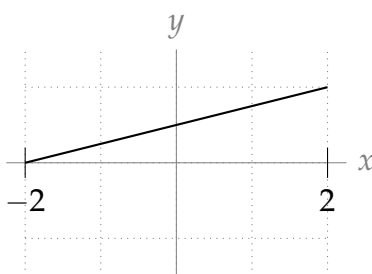
(b)



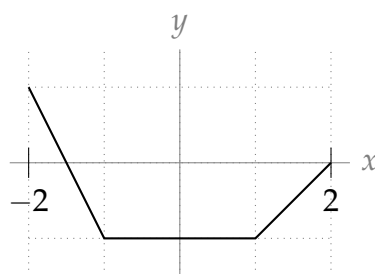
(c)



(d)



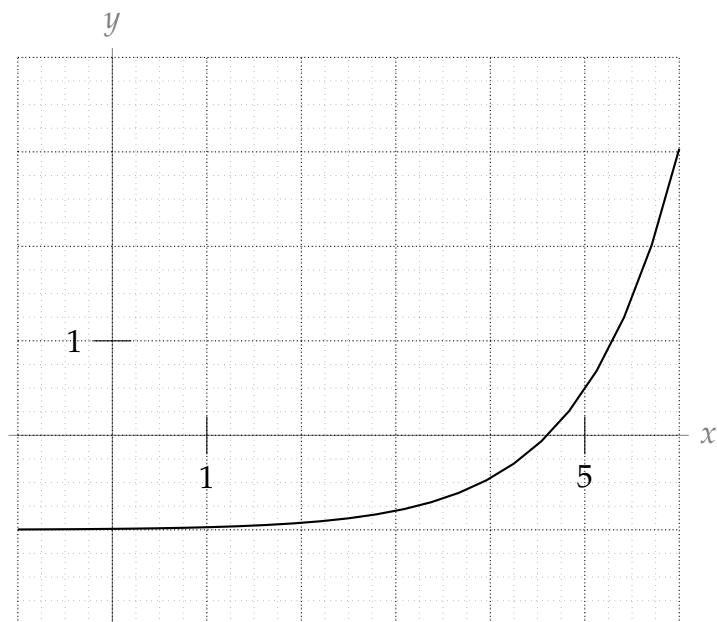
(e)



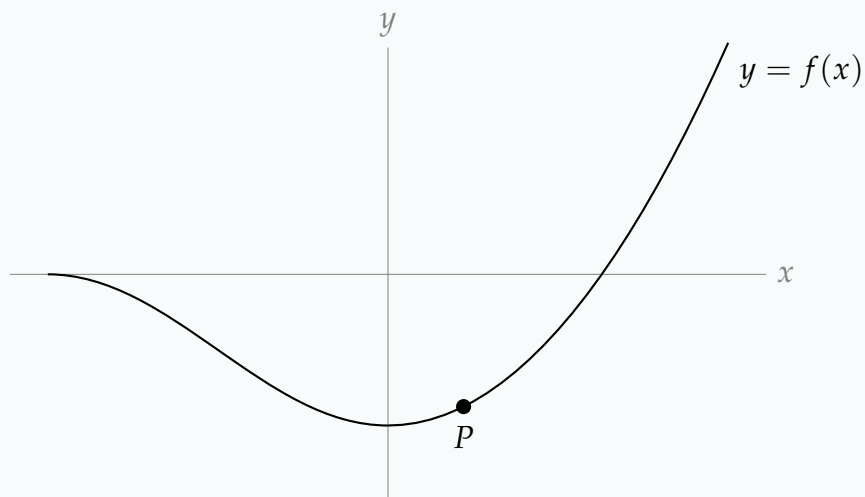
(f)

## ► Stage 2

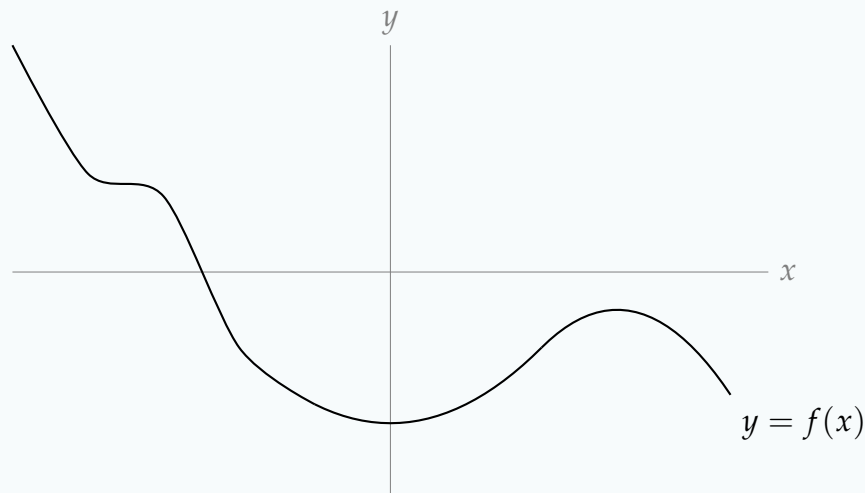
Q[4]: Give your best approximation of the slope of the tangent line to the graph below at the point  $x = 5$ .



Q[5]: On the graph below, sketch the tangent line to  $y = f(x)$  at  $P$ . Then, find two points  $Q$  and  $R$  on the graph so that the secant line through  $Q$  and  $R$  has the same slope as the tangent line at  $P$ .



Q[6]: Mark the points where the curve shown below has a tangent line with slope 0.



(Later on, we'll learn how these points tell us a lot about the shape of a graph.)

## 2.2▲ Definition of the derivative

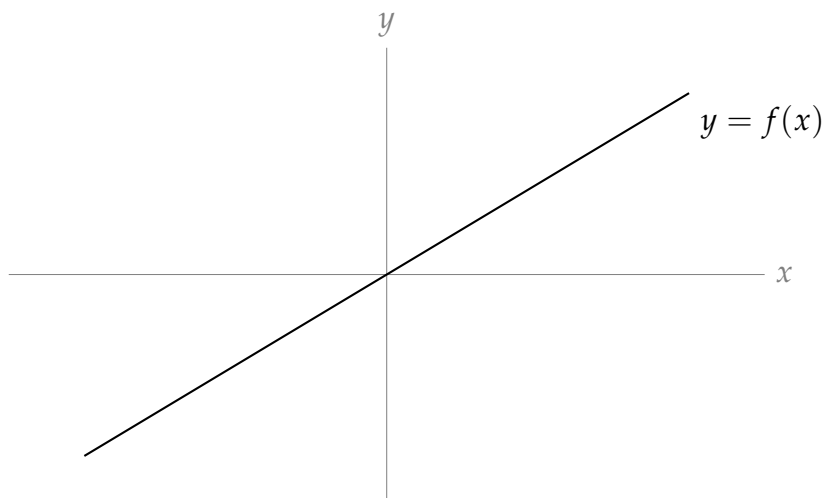
### Exercises

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#### ► Stage 1

Q[1]: The function  $f(x)$  is shown. Select all options below that describe its derivative,  $\frac{df}{dx}$ :

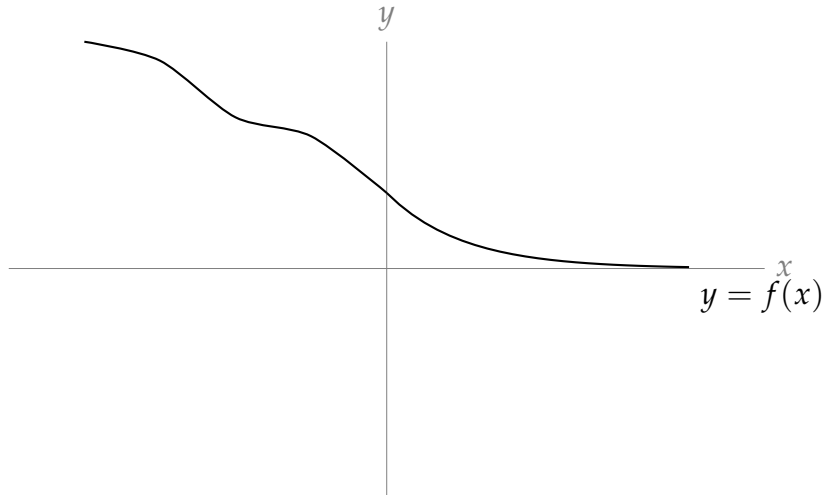
- (a) constant      (b) increasing      (c) decreasing  
(d) always positive      (e) always negative



Q[2]: The function  $f(x)$  is shown. Select all options below that describe its derivative,  $\frac{df}{dx}$ :

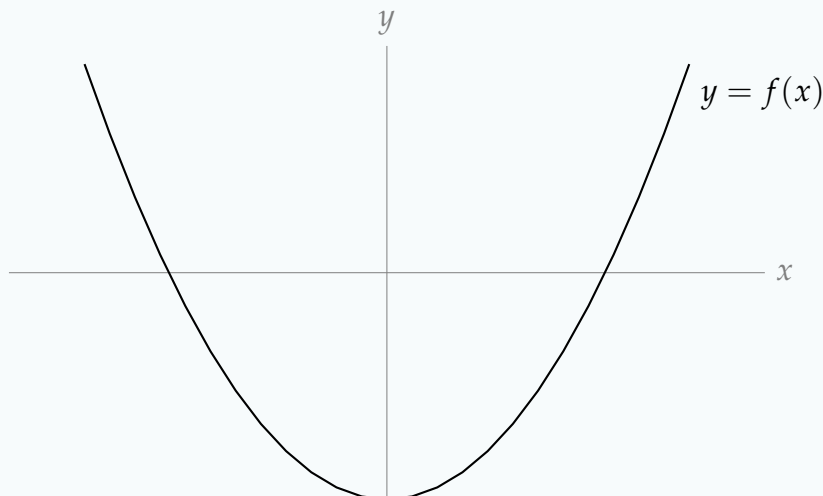
- (a) constant      (b) increasing      (c) decreasing

(d) always positive      (e) always negative



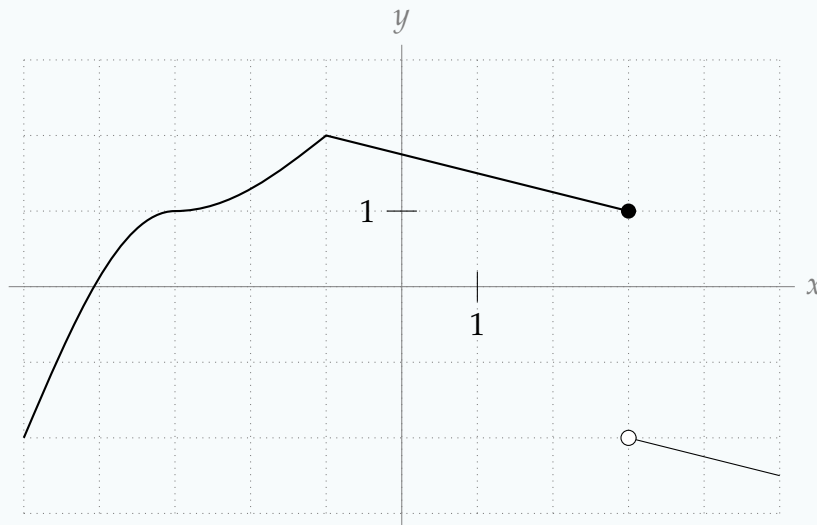
Q[3]: The function  $f(x)$  is shown. Select all options below that describe its derivative,  $\frac{df}{dx}$ :

- (a) constant      (b) increasing      (c) decreasing  
 (d) always positive      (e) always negative



Q[4](\*) : State, in terms of a limit, what it means for  $f(x) = x^3$  to be differentiable at  $x = 0$ .

Q[5]: For which values of  $x$  does  $f'(x)$  not exist?



Q[6]: Suppose  $f(x)$  is a function defined at  $x = a$  with

$$\lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} = 1.$$

True or false:  $f'(a) = 1$ .

Q[7]: Suppose  $f(x)$  is a function defined at  $x = a$  with

$$\lim_{x \rightarrow a^-} f'(x) = \lim_{x \rightarrow a^+} f'(x) = 1.$$

True or false:  $f'(a) = 1$ .

Q[8]: Suppose  $s(t)$  is a function, with  $t$  measured in seconds, and  $s$  measured in metres. What are the units of  $s'(t)$ ?

## ► Stage 2

Q[9]: Use the definition of the derivative to find the equation of the tangent line to the curve  $y(x) = x^3 + 5$  at the point  $(1, 6)$ .

Q[10]: Use the definition of the derivative to find the derivative of  $f(x) = \frac{1}{x}$ .

Q[11](\*) : Let  $f(x) = x|x|$ . Using the definition of the derivative, show that  $f(x)$  is differentiable at  $x = 0$ .

Q[12](\*) : Use the definition of the derivative to compute the derivative of the function  $f(x) = \frac{2}{x+1}$ .

Q[13](\*) : Use the definition of the derivative to compute the derivative of the function  $f(x) = \frac{1}{x^2+3}$ .

Q[14]: Use the definition of the derivative to find the slope of the tangent line to the curve  $f(x) = x \log_{10}(2x + 10)$  at the point  $x = 0$ .

Q[15](\*) : Compute the derivative of  $f(x) = \frac{1}{x^2}$  directly from the definition.

Q[16](\*) : Find the values of the constants  $a$  and  $b$  for which

$$f(x) = \begin{cases} x^2 & x \leq 2 \\ ax + b & x > 2 \end{cases}$$

is differentiable everywhere.

Remark: In the text, you have already learned the derivatives of  $x^2$  and  $ax + b$ . In this question, you are only asked to find the values of  $a$  and  $b$ —not to justify how you got them—so you don't have to use the definition of the derivative. However, on an exam, you might be asked to justify your answer, in which case you would show how to differentiate the two branches of  $f(x)$  using the definition of a derivative.

Q[17](\*) : Use the definition of the derivative to compute  $f'(x)$  if  $f(x) = \sqrt{1+x}$ . Where does  $f'(x)$  exist?

### ► Stage 3

Q[18]: Use the definition of the derivative to find the velocity of an object whose position is given by the function  $s(t) = t^4 - t^2$ .

Q[19](\*) : Determine whether the derivative of following function exists at  $x = 0$ .

$$f(x) = \begin{cases} x \cos x & \text{if } x \geq 0 \\ \sqrt{x^2 + x^4} & \text{if } x < 0 \end{cases}$$

You must justify your answer using the definition of a derivative.

Q[20](\*) : Determine whether the derivative of the following function exists at  $x = 0$

$$f(x) = \begin{cases} x \cos x & \text{if } x \leq 0 \\ \sqrt{1+x} - 1 & \text{if } x > 0 \end{cases}$$

You must justify your answer using the definition of a derivative.

Q[21](\*) : Determine whether the derivative of the following function exists at  $x = 0$

$$f(x) = \begin{cases} x^3 - 7x^2 & \text{if } x \leq 0 \\ x^3 \cos\left(\frac{1}{x}\right) & \text{if } x > 0 \end{cases}$$

You must justify your answer using the definition of a derivative.

Q[22](\*) : Determine whether the derivative of the following function exists at  $x = 1$

$$f(x) = \begin{cases} 4x^2 - 8x + 4 & \text{if } x \leq 1 \\ (x-1)^2 \sin\left(\frac{1}{x-1}\right) & \text{if } x > 1 \end{cases}$$

You must justify your answer using the definition of a derivative.

Q[23]: Sketch a function  $f(x)$  with  $f'(0) = -1$  that takes the following values:

|             |    |                |                |                |   |               |               |               |   |
|-------------|----|----------------|----------------|----------------|---|---------------|---------------|---------------|---|
| <b>x</b>    | -1 | $-\frac{1}{2}$ | $-\frac{1}{4}$ | $-\frac{1}{8}$ | 0 | $\frac{1}{8}$ | $\frac{1}{4}$ | $\frac{1}{2}$ | 1 |
| <b>f(x)</b> | -1 | $-\frac{1}{2}$ | $-\frac{1}{4}$ | $-\frac{1}{8}$ | 0 | $\frac{1}{8}$ | $\frac{1}{4}$ | $\frac{1}{2}$ | 1 |

Remark: you can't always guess the behaviour of a function from its points, even if the points seem to be making a clear pattern.

Q[24]: Let  $p(x) = f(x) + g(x)$ , for some functions  $f$  and  $g$  whose derivatives exist. Use limit laws and the definition of a derivative to show that  $p'(x) = f'(x) + g'(x)$ .

Remark: this is called the sum rule, and we'll learn more about it in Lemma 2.4.1.

Q[25]: Let  $f(x) = 2x$ ,  $g(x) = x$ , and  $p(x) = f(x) \cdot g(x)$ .

- Find  $f'(x)$  and  $g'(x)$ .
- Find  $p'(x)$ .
- Is  $p'(x) = f'(x) \cdot g'(x)$ ?

In Theorem 2.4.3, you'll learn a rule for calculating the derivative of a product of two functions.

Q[26](\*) : There are two distinct straight lines that pass through the point  $(1, -3)$  and are tangent to the curve  $y = x^2$ . Find equations for these two lines.

Remark: the point  $(1, -3)$  does not lie on the curve  $y = x^2$ .

Q[27](\*) : For which values of  $a$  is the function

$$f(x) = \begin{cases} 0 & x \leq 0 \\ x^a \sin \frac{1}{x} & x > 0 \end{cases}$$

differentiable at 0?

## 2.3▲ Interpretations of the derivative

### Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

- Let's start with integers.  $1^3 < 7 < 2^3$ , so  $\sqrt[3]{7}$  is in the interval  $(1, 2)$ .
- Let's narrow this down, say by testing  $f(1.5)$ .  $(1.5)^3 = 3.375 < 7$ , so  $\sqrt[3]{7}$  is in the interval  $(1.5, 2)$ .
- Let's narrow further, say by testing  $f(1.75)$ .  $(1.75)^3 \approx 5.34 < 7$ , so  $\sqrt[3]{7}$  is in the interval  $(1.75, 2)$ .
- Testing various points, we find  $f(1.9) < 7 < f(2)$ , so  $\sqrt[3]{7}$  is between 1.9 and 2.
- By testing more, we find  $f(1.91) < 7 < f(1.92)$ , so  $\sqrt[3]{7}$  is in  $(1.91, 1.92)$ .
- In order to get an approximation for  $\sqrt[3]{7}$  that is rounded to two decimal places, we have to know whether  $\sqrt[3]{7}$  is greater or less than 1.915; indeed  $f(1.915) \approx 7.02 > 7$ , so  $\sqrt[3]{7} < 1.915$ ; then rounded to two decimal places,  $\sqrt[3]{7} \approx 1.91$ .

If this seems like the obvious way to approximate  $\sqrt[3]{7}$ , that's good. The IVT is a formal statement of a very intuitive principle.

**S-28:**

- If  $f(a) = g(a)$ , or  $f(b) = g(b)$ , then we simply take  $c = a$  or  $c = b$ .
- Suppose  $f(a) \neq g(a)$  and  $f(b) \neq g(b)$ . Then  $f(a) < g(a)$  and  $g(b) < f(b)$ , so if we define  $h(x) = f(x) - g(x)$ , then  $h(a) < 0$  and  $h(b) > 0$ . Since  $h$  is the difference of two functions that are continuous over  $[a, b]$ , also  $h$  is continuous over  $[a, b]$ . So, by the Intermediate Value Theorem, there exists some  $c \in (a, b)$  with  $h(c) = 0$ ; that is,  $f(c) = g(c)$ .

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## Solutions to Exercises **2.1** — Jump to [TABLE OF CONTENTS](#)

**S-1:** If  $Q$  is to the left of the  $y$  axis, the line through  $Q$  and  $P$  is increasing, so the secant line has positive slope. If  $Q$  is to the right of the  $y$  axis, the line through  $Q$  and  $P$  is decreasing, so the secant line has negative slope.

**S-2:** (a) By drawing a few pictures, it's easy to see that sliding  $Q$  closer to  $P$ , the slope of the secant line increases.

(b) Since the slope of the secant line increases the closer  $Q$  gets to  $P$ , that means the tangent line (which is the limit as  $Q$  approaches  $P$ ) has a larger slope than the secant line between  $Q$  and  $P$  (using the location where  $Q$  is right now).

Alternately, by simply sketching the tangent line at  $P$ , we can see that has a steeper slope than the secant line between  $P$  and  $Q$ .

**S-3:** The slope of the secant line will be  $\frac{f(2) - f(-2)}{2 - (-2)} = \frac{f(2) - f(-2)}{4}$ , in every part. So, if two lines have the same slope, that means their differences  $f(2) - f(-2)$  will be the same.

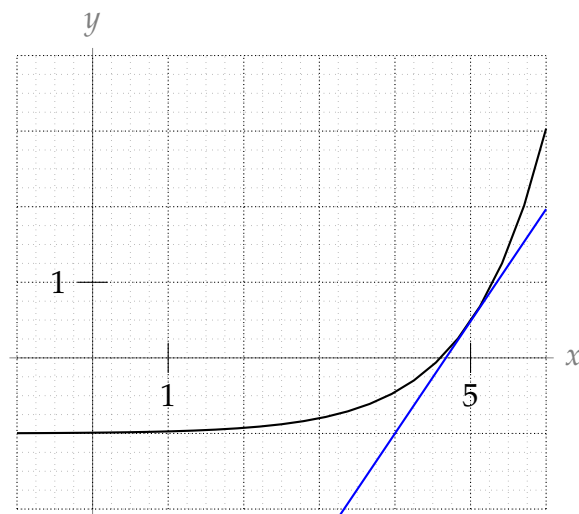
The graphs in (a),(c), and (e) all have  $f(2) - f(-2) = 1$ , so they all have the same secant line slope. The graphs in (b) and (f) both have  $f(2) - f(-2) = -1$ , so they both have the same secant line slope. The graph in (d) has  $f(2) - f(-2) = 0$ , and it is the only graph with this property, so it does not share its secant line slope with any of the other graphs.

**S-4:** A good approximation from the graph is  $f(5) = 0.5$ . We want to find a secant line whose endpoints are both very close to  $x = 5$ , but that also give us clear  $y$ -values. It looks like  $f(5.25) \approx 1$ , and  $f(4.75) \approx \frac{1}{8}$ . The secant line from  $x = 5$  to  $x = 5.25$  has approximate slope  $\frac{f(5.25) - f(5)}{5.25 - 5} \approx \frac{1 - .5}{.25} = 2$ . The secant line from  $x = 5$  to  $x = 4.75$  has approximate slope  $\frac{0.5 - \frac{1}{8}}{5 - 4.75} = \frac{3}{2}$ .

The graph increases more and more quickly (gets steeper and steeper), so it makes sense that the secant line to the left of  $x = 5$  has a smaller slope than the secant line to the right of  $x = 5$ . Also, if you're taking secant lines that have endpoints farther out from  $x = 5$ , you'll notice that the slopes of the secant lines change quite dramatically. You have to be very, very close to  $x = 5$  to get any kind of accuracy.

If we split the difference, we might approximate the slope of the secant line to be the average of  $\frac{3}{2}$  and 2, which is  $\frac{7}{4}$ .

Another way to try to figure out the tangent line is by carefully drawing it in with a ruler. This is shown here in blue:

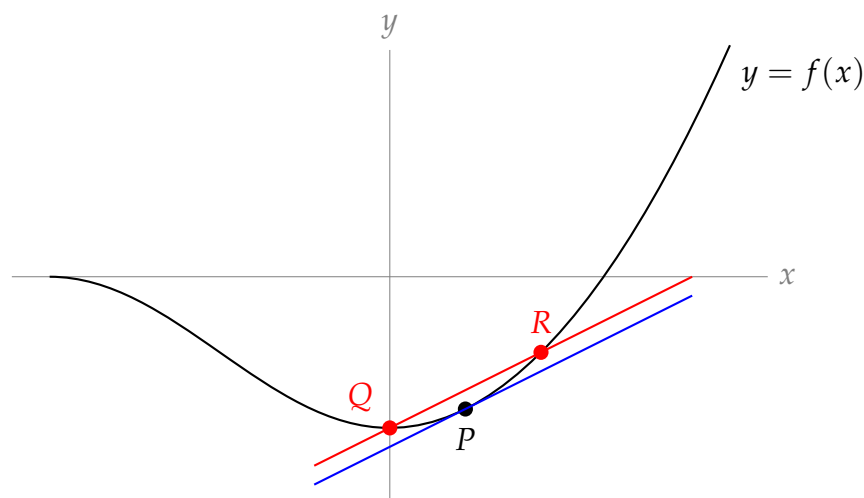


It's much easier to take the slope of a line than a curve, and this one looks like it has slope about 1.5. However, we drew this with a computer: by hand it's much harder to draw an accurate tangent line. (That's why we need calculus!)

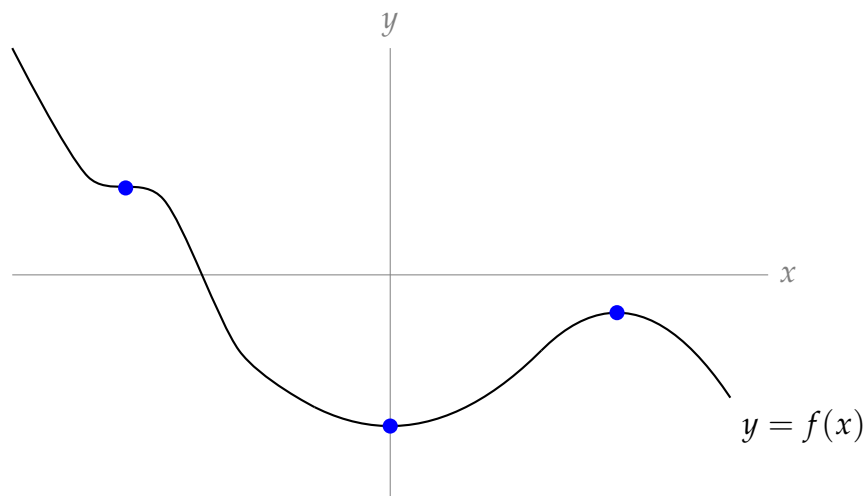
The actual slope of the tangent line to the function at  $x = 5$  is about 1.484. This is extremely hard to figure out just from the graph—by hand, a guess between 1.25 and 1.75 would be very accurate.

**S-5:** There is only one tangent line to  $f(x)$  at  $P$  (shown in blue), but there are infinitely many choices of  $Q$  and  $R$  (one possibility shown in red). One easy way to sketch the

secant line on paper is to draw any line parallel to the tangent line, and choose two intercepts with  $y = f(x)$ .



**S-6:** Any place the graph looks flat (if you imagine zooming in) is where the tangent line has slope 0. This occurs three times.



Notice that two of the indicated points are at a low point and a high point, respectively. Later, we'll use these places where the tangent line has slope zero to find where a graph achieves its biggest and smallest values.

**Solutions to Exercises 2.2** — Jump to [TABLE OF CONTENTS](#)

**S-1:** The function shown is a line, so it has a constant slope—(a) . Since the function is always increasing,  $f'$  is always positive, so also (d) holds. Remark: it does not matter that the function itself is sometimes negative; the slope is always positive because the function is always increasing. Also, since the slope is constant,  $f'$  is neither increasing nor decreasing: it is the *function* that is increasing, not its derivative.

**S-2:** The function is always decreasing, so  $f'$  is always negative, option (e). However, the function alternates between being more and less steep, so  $f'$  alternates between increasing and decreasing several times, and no other options hold.

Remark:  $f$  is always positive, but (d) does not hold!

**S-3:** At the left end of the graph,  $f$  is decreasing rapidly, so  $f'$  is a strongly negative number. Then as we move towards  $x = 0$ ,  $f$  decreases less rapidly, so  $f'$  is a less strongly negative number. As we pass 0,  $f$  increases, so  $f'$  is a positive number. As we move to the right,  $f$  increases more and more rapidly, so  $f'$  is an increasing positive number. This description tells us that  $f'$  increases for the entire range shown. So (b) holds, but not (a) or (c). Since  $f'$  is negative to the left of the  $y$  axis, and positive to the right of it, also (d) and (e) do not hold.

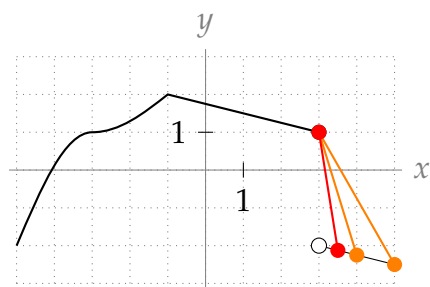
**S-4:** By definition,  $f(x) = x^3$  is differentiable at  $x = 0$  if the limit

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^3 - 0}{h}$$

exists.

**S-5:**  $f'(-1)$  does not exist, because to the left of  $x = -1$  the slope is a pretty big positive number (looks like around  $+1$ ) and to the right the slope is  $-1/4$ . Since the derivative involves a limit, that limit needs to match the limit from the left and the limit from the right. The sharp angle made by the graph at  $x = -1$  indicates that the left and right limits do not match, so the derivative does not exist.

$f'(3)$  also does not exist. One way to see this is to notice that the function is discontinuous here. More viscerally, note that  $f(3) = 1$ , so as we take secant lines with one endpoint  $(3, 1)$ , and the other endpoint just to the right of  $x = 3$ , we get slopes that are more and more strongly negative, as shown in the picture below. If we take the limit of the slopes of these secant lines as  $x$  goes to 3 from the right, we get  $-\infty$ . (This certainly doesn't match the slope from the left, which is  $-\frac{1}{4}$ .)



At  $x = -3$ , there is some kind of “change” in the graph; however, it is a smooth curve, so the derivative exists here.

**S-6:** True. The definition of the derivative tells us that

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

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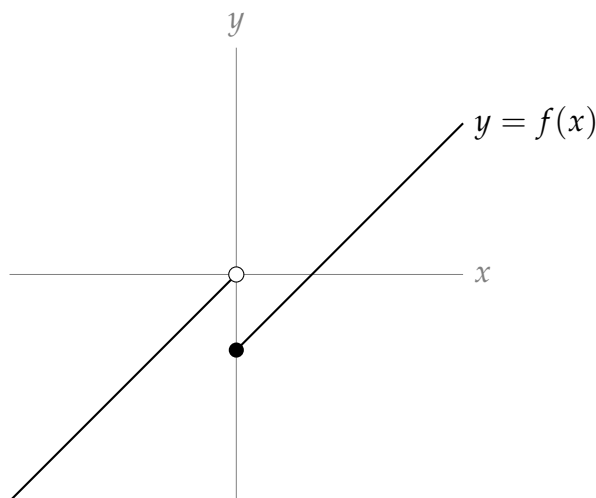
if it exists. We know from our work with limits that if both one-sided limits  $\lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}$  and  $\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$  exist and are equal to each other, then  $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$  exists and has the same value as the one-sided limits. So, since the one-sided limits exist and are equal to one, we conclude  $f'(a)$  exists and is equal to one.

**S-7:** In general, this is false. The key problem that can arise is that  $f(x)$  might not be continuous at  $x = 1$ . One example is the function

$$f(x) = \begin{cases} x & x < 0 \\ x - 1 & x \geq 0 \end{cases}$$

where  $f'(x) = 1$  whenever  $x \neq 0$  (so in particular,  $\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^+} f'(x) = 1$ ) but  $f'(0)$  does not exist.

There are two ways to see that  $f'(0)$  does not exist. One is to notice that  $f$  is not continuous at  $x = 0$ .



Another way to see that  $f'(0)$  does not exist is to use the definition of the derivative. Remember, in order for a limit to exist, both one-sided limits must exist. Let's consider the limit from the left. If  $h \rightarrow 0^-$ , then  $h < 0$ , so  $f(h)$  is equal to  $h$  (not  $h - 1$ ).

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^-} \frac{(h) - (-1)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{h+1}{h} \\ &= \lim_{h \rightarrow 0^-} 1 + \frac{1}{h} \\ &= -\infty \end{aligned}$$

In particular, this limit does not exist. Since the one-sided limit does not exist,

$$\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = DNE$$

---

and so  $f'(0)$  does not exist.

**S-8:** Using the definition of the derivative,

$$s'(t) = \lim_{h \rightarrow 0} \frac{s(t+h) - s(t)}{h}$$

The units of the numerator are meters, and the units of the denominator are seconds (since the denominator comes from the change in the *input* of the function). So, the units of  $s'(t)$  are metres per second.

Remark: we learned that the derivative of a position function gives velocity. In this example, the position is given in metres, and the velocity is measured in metres per second.

**S-9:** We can use point-slope form to get the equation of the line, if we have a point and its slope. The point is given:  $(1, 6)$ . The slope is the derivative:

$$\begin{aligned} y'(1) &= \lim_{h \rightarrow 0} \frac{y(1+h) - y(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(1+h)^3 + 5] - [1^3 + 5]}{h} \\ &= \lim_{h \rightarrow 0} \frac{[1 + 3h + 3h^2 + h^3 + 5] - [1 + 5]}{h} \\ &= \lim_{h \rightarrow 0} \frac{3h + 3h^2 + h^3}{h} \\ &= \lim_{h \rightarrow 0} 3 + 3h + h^2 \\ &= 3 \end{aligned}$$

So our slope is 3, which gives a line of equation  $y - 6 = 3(x - 1)$ .

---

**S-10:** We set up the definition of the derivative.

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\&= \lim_{h \rightarrow 0} \frac{\frac{x}{x(x+h)} - \frac{x+h}{x(x+h)}}{h} \\&= \lim_{h \rightarrow 0} \frac{\frac{x-(x+h)}{x(x+h)}}{h} \\&= \lim_{h \rightarrow 0} \frac{\frac{-h}{x(x+h)}}{h} \\&= \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} \\&= \frac{-1}{x^2}\end{aligned}$$

**S-11:** By definition

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h|h|}{h} = \lim_{h \rightarrow 0} |h| = 0$$

In particular, the limit exists, so the derivative exists (and is equal to zero).

**S-12:** We set up the definition of the derivative.

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left( \frac{2}{x+h+1} - \frac{2}{x+1} \right) = \lim_{h \rightarrow 0} \frac{2}{h} \frac{(x+1) - (x+h+1)}{(x+h+1)(x+1)} \\&= \lim_{h \rightarrow 0} \frac{2}{h} \frac{-h}{(x+h+1)(x+1)} = \lim_{h \rightarrow 0} \frac{-2}{(x+h+1)(x+1)} = \frac{-2}{(x+1)^2}\end{aligned}$$

**S-13:**

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left( \frac{1}{(x+h)^2 + 3} - \frac{1}{x^2 + 3} \right) = \lim_{h \rightarrow 0} \frac{1}{h} \frac{x^2 - (x+h)^2}{[(x+h)^2 + 3][x^2 + 3]} \\&= \lim_{h \rightarrow 0} \frac{1}{h} \frac{-2xh - h^2}{[(x+h)^2 + 3][x^2 + 3]} = \lim_{h \rightarrow 0} \frac{-2x - h}{[(x+h)^2 + 3][x^2 + 3]} = \boxed{\frac{-2x}{[x^2 + 3]^2}}\end{aligned}$$

**S-14:** The slope of the tangent line is the derivative. We set this up using the same definition of the derivative that we always do. This limit is hard to take for general  $x$ , but

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easy when  $x = 0$ .

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h \log_{10}(2h+10) - 0}{h} \\ &= \lim_{h \rightarrow 0} \log_{10}(2h+10) = \log_{10}(10) = 1 \end{aligned}$$

So, the slope of the tangent line is 1.

**S-15:**

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h} = \lim_{h \rightarrow 0} \frac{x^2 - (x+h)^2}{(x+h)^2 x^2 h} = \lim_{h \rightarrow 0} \frac{-2xh - h^2}{(x+h)^2 x^2 h} \\ &= \lim_{h \rightarrow 0} \frac{-2x - h}{(x+h)^2 x^2} = \frac{-2x}{x^4} = -\frac{2}{x^3} \end{aligned}$$

**S-16:** When  $x$  is not equal to 2, then the function is differentiable— the only place we have to worry about is when  $x$  is exactly 2.

In order for  $f$  to be differentiable at  $x = 2$ , it must also be continuous at  $x = 2$ . This forces  $x^2|_{x=2} = [ax + b]_{x=2}$  or

$$2a + b = 4.$$

In order for a limit to exist, the left- and right-hand limits must exist and be equal to each other. Since a derivative is a limit, in order for  $f$  to be differentiable at  $x = 2$ , the left hand derivative of  $ax + b$  at  $x = 2$  must be the same as the right hand derivative of  $x^2$  at  $x = 2$ . Since  $ax + b$  is a line, its derivative is  $a$  everywhere. We've already seen the derivative of  $x^2$  is  $2x$ , so we need

$$a = 2x|_{x=2} = 4.$$

So, the values of  $a$  and  $b$  that makes  $f$  differentiable everywhere are  $a = 4$  and  $b = -4$ .

**S-17:** We plug in  $f(x)$  to the definition of a derivative. To evaluate the limit, we multiply

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and divide by the conjugate of the numerator, then simplify.

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{1+x+h} - \sqrt{1+x}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{1+x+h} - \sqrt{1+x}}{h} \left( \frac{\sqrt{1+x+h} + \sqrt{1+x}}{\sqrt{1+x+h} + \sqrt{1+x}} \right) \\
 &= \lim_{h \rightarrow 0} \frac{(1+x+h) - (1+x)}{h(\sqrt{1+x+h} + \sqrt{1+x})} \\
 &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{1+x+h} + \sqrt{1+x})} \\
 &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{1+x+h} + \sqrt{1+x}} \\
 &= \frac{1}{\sqrt{1+x+0} + \sqrt{1+x}} = \frac{1}{2\sqrt{1+x}}
 \end{aligned}$$

The domain of the function is  $[-1, \infty)$ . In particular,  $f(x)$  is defined when  $x = -1$ . However,  $f'(x)$  is not defined when  $x = -1$ , so  $f'(x)$  only exists over  $(-1, \infty)$ .

Remark:  $\lim_{x \rightarrow -1^+} f'(x) = \infty$ , so the tangent line to  $f(x)$  at the point  $x = -1$  has a vertical slope.

**S-18:** From Section 1.2, we see that the velocity is exactly the derivative.

$$\begin{aligned}
 v(t) &= \lim_{h \rightarrow 0} \frac{s(t+h) - s(t)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(t+h)^4 - (t+h)^2 - t^4 + t^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(t^4 + 4t^3h + 6t^2h^2 + 4th^3 + h^4) - (t^2 + 2th + h^2) - t^4 + t^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4t^3h + 6t^2h^2 + 4th^3 + h^4 - 2th - h^2}{h} \\
 &= \lim_{h \rightarrow 0} 4t^3 + 6t^2h + 4th^2 + h^3 - 2t - h \\
 &= 4t^3 - 2t
 \end{aligned}$$

So, the velocity is given by  $v(t) = 4t^3 - 2t$ .

**S-19:** The function is differentiable at  $x = 0$  if the following limit:

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x) - 0}{x} = \lim_{x \rightarrow 0} \frac{f(x)}{x}$$

exists (note that we used the fact that  $f(0) = 0$  as per the definition of the first branch which includes the point  $x = 0$ ). We start by computing the left limit. For this computation, recall that if  $x < 0$  then  $\sqrt{x^2} = |x| = -x$ .

$$\lim_{x \rightarrow 0^-} \frac{f(x)}{x} = \lim_{x \rightarrow 0^-} \frac{\sqrt{x^2 + x^4}}{x} = \lim_{x \rightarrow 0^-} \frac{\sqrt{x^2} \sqrt{1 + x^2}}{x} = \lim_{x \rightarrow 0^-} \frac{-x \sqrt{1 + x^2}}{x} = -1$$

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Now, from the right:

$$\lim_{x \rightarrow 0^+} \frac{x \cos x}{x} = \lim_{x \rightarrow 0^+} \cos x = 1.$$

Since the limit from the left does not equal the limit from the right, the derivative does not exist at  $x = 0$ .

**S-20:** The function is differentiable at  $x = 0$  if the following limit:

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x) - 0}{x} = \lim_{x \rightarrow 0} \frac{f(x)}{x}$$

exists (note that we used the fact that  $f(0) = 0$  as per the definition of the first branch which includes the point  $x = 0$ ).

We start by computing the left limit.

$$\lim_{x \rightarrow 0^-} \frac{f(x)}{x} = \lim_{x \rightarrow 0^-} \frac{x \cos x}{x} = \lim_{x \rightarrow 0^-} \cos x = 1.$$

Now, from the right:

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{\sqrt{1+x} - 1}{x} &= \lim_{x \rightarrow 0^+} \frac{\sqrt{1+x} - 1}{x} \cdot \frac{\sqrt{1+x} + 1}{\sqrt{1+x} + 1} \\ &= \lim_{x \rightarrow 0^+} \frac{1 + x - 1}{x(\sqrt{1+x} + 1)} = \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{1+x} + 1} = \frac{1}{2} \end{aligned}$$

Since the limit from the left does not equal the limit from the right, the derivative does not exist at  $x = 0$ .

**S-21:** The function is differentiable at  $x = 0$  if the following limit:

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x) - 0}{x} = \lim_{x \rightarrow 0} \frac{f(x)}{x}$$

exists (note that we used the fact that  $f(0) = 0$  as per the definition of the first branch which includes the point  $x = 0$ ). We compute left and right limits; so

$$\lim_{x \rightarrow 0^-} \frac{f(x)}{x} = \lim_{x \rightarrow 0^-} \frac{x^3 - 7x^2}{x} = \lim_{x \rightarrow 0^-} x^2 - 7x = 0$$

and

$$\lim_{x \rightarrow 0^+} \frac{x^3 \cos\left(\frac{1}{x}\right)}{x} = \lim_{x \rightarrow 0^+} x^2 \cdot \cos\left(\frac{1}{x}\right).$$

This last limit equals 0 by the Squeeze Theorem since

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$$

---

and so,

$$-x^2 \leq x^2 \cdot \cos\left(\frac{1}{x}\right) \leq x^2,$$

where in these inequalities we used the fact that  $x^2 \geq 0$ . Finally, since  $\lim_{x \rightarrow 0^+} -x^2 = \lim_{x \rightarrow 0^+} x^2 = 0$ , the Squeeze Theorem yields that also  $\lim_{x \rightarrow 0^+} x^2 \cos\left(\frac{1}{x}\right) = 0$ , as claimed.

Since the left and right limits match (they're both equal to 0), we conclude that indeed  $f(x)$  is differentiable at  $x = 0$  (and its derivative at  $x = 0$  is actually equal to 0).

**S-22:** The function is differentiable at  $x = 1$  if the following limit:

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{f(x) - 0}{x - 1} = \lim_{x \rightarrow 1} \frac{f(x)}{x - 1}$$

exists (note that we used the fact that  $f(1) = 0$  as per the definition of the first branch which includes the point  $x = 0$ ). We compute left and right limits; so

$$\lim_{x \rightarrow 1^-} \frac{f(x)}{x - 1} = \lim_{x \rightarrow 1^-} \frac{4x^2 - 8x + 4}{x - 1} = \lim_{x \rightarrow 1^-} \frac{4(x - 1)^2}{x - 1} = \lim_{x \rightarrow 1^-} 4(x - 1) = 0$$

and

$$\lim_{x \rightarrow 1^+} \frac{(x - 1)^2 \sin\left(\frac{1}{x - 1}\right)}{x - 1} = \lim_{x \rightarrow 1^+} (x - 1) \cdot \sin\left(\frac{1}{x - 1}\right).$$

This last limit equals 0 by the Squeeze Theorem since

$$-1 \leq \sin\left(\frac{1}{x - 1}\right) \leq 1$$

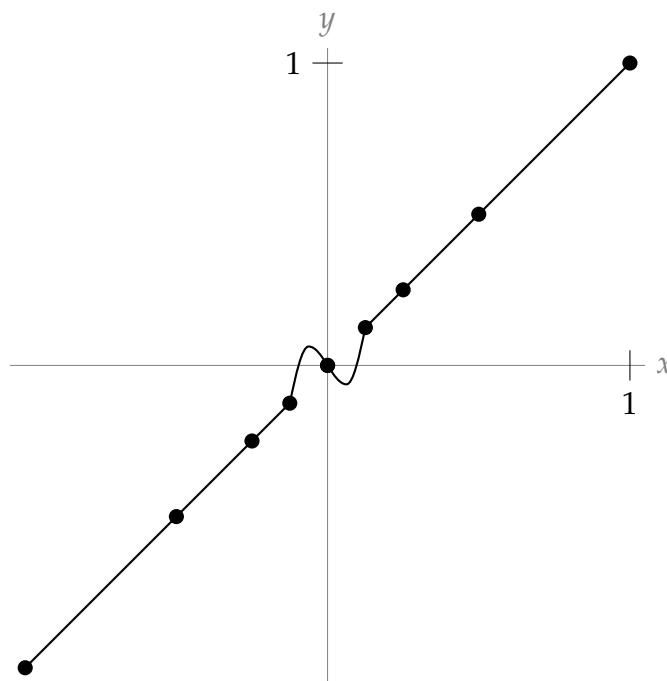
and so,

$$-(x - 1) \leq (x - 1) \cdot \sin\left(\frac{1}{x - 1}\right) \leq x - 1,$$

where in these inequalities we used the fact that  $x \rightarrow 1^+$  yields positive values for  $x - 1$ . Finally, since  $\lim_{x \rightarrow 1^+} -x + 1 = \lim_{x \rightarrow 1^+} x - 1 = 0$ , the Squeeze Theorem yields that also  $\lim_{x \rightarrow 1^+} (x - 1) \sin\left(\frac{1}{x - 1}\right) = 0$ , as claimed.

Since the left and right limits match (they're both equal to 0), we conclude that indeed  $f(x)$  is differentiable at  $x = 1$  (and its derivative at  $x = 1$  is actually equal to 0).

**S-23:** Many answers are possible; here is one.



The key is to realize that the few points you're given suggest a pattern, but don't guarantee it. You only know nine points; anything can happen in between.

**S-24:**

$$\begin{aligned}
 p'(x) &= \lim_{h \rightarrow 0} \frac{p(x+h) - p(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h) + g(x+h) - f(x) - g(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x) + g(x+h) - g(x)}{h} \\
 &= \lim_{h \rightarrow 0} \left[ \frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right] \\
 (*) &= \left[ \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \right] + \left[ \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \right] \\
 &= f'(x) + g'(x)
 \end{aligned}$$

At step (\*), we use the limit law that  $\lim_{x \rightarrow a} [F(x) + G(x)] = \lim_{x \rightarrow a} F(x) + \lim_{x \rightarrow a} G(x)$ , as long as  $\lim_{x \rightarrow a} F(x)$  and  $\lim_{x \rightarrow a} G(x)$  exist. Because the problem states that  $f'(x)$  and  $g'(x)$  exist, we know that  $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$  and  $\lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$  exist, so our work is valid.

**S-25:** (a) Since  $y = f(x) = 2x$  and  $y = g(x) = x$  are straight lines, we don't need the definition of the derivative (although you can use it if you like).  $f'(x) = 2$  and  $g'(x) = 1$ .

---

(b)  $p(x) = 2x^2$ , so  $p(x)$  is not a line: we use the definition of a derivative to find  $p'(x)$ .

$$\begin{aligned} p'(x) &= \lim_{h \rightarrow 0} \frac{p(x+h) - p(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2(x+h)^2 - 2x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 2x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{h} \\ &= \lim_{h \rightarrow 0} 4x + 2h \\ &= 4x \end{aligned}$$

(c) No,  $p'(x) = 4x \neq 2 \cdot 1 = f'(x) \cdot g'(x)$ . In general, the derivative of a product is not the same as the derivative of the functions being multiplied.

S-26: We know that  $y' = 2x$ . So, if we choose a point  $(\alpha, \alpha^2)$  on the curve  $y = x^2$ , then the tangent line to the curve at that point has slope  $2\alpha$ . That is, the tangent line has equation

$$\begin{aligned} (y - \alpha^2) &= 2\alpha(x - \alpha) \\ \text{simplified, } y &= (2\alpha)x - \alpha^2 \end{aligned}$$

So, if  $(1, -3)$  is on the tangent line, then

$$\begin{aligned} -3 &= (2\alpha)(1) - \alpha^2 \\ \iff 0 &= \alpha^2 - 2\alpha - 3 \\ \iff 0 &= (\alpha - 3)(\alpha + 1) \\ \iff \alpha &= 3, \quad \text{or} \quad \alpha = -1. \end{aligned}$$

So, the tangent lines  $y = (2\alpha)x - \alpha^2$  are

$$y = 6x - 9 \quad \text{and} \quad y = -2x - 1.$$

S-27: Using the definition of the derivative,  $f$  is differentiable at 0 if and only if

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

exists. In particular, this means  $f$  is differentiable at 0 if and only if both one-sided limits exist and are equal to each other.

When  $h < 0$ ,  $f(h) = 0$ , so

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{0 - 0}{h} = 0$$

---

So,  $f$  is differentiable at  $x = 0$  if and only if

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = 0.$$

To evaluate the limit above, we note  $f(0) = 0$  and, when  $h > 0$ ,  $f(h) = h^a \sin\left(\frac{1}{h}\right)$ , so

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} \frac{h^a \sin\left(\frac{1}{h}\right)}{h} \\ &= \lim_{h \rightarrow 0^+} h^{a-1} \sin\left(\frac{1}{h}\right) \end{aligned}$$

We will spend the rest of this solution evaluating the limit above for different values of  $a$ , to find when it is equal to zero and when it is not. Let's consider the different values that could be taken by  $h^{a-1}$ .

- If  $a = 1$ , then  $a - 1 = 0$ , so  $h^{a-1} = h^0 = 1$  for all values of  $h$ . Then

$$\lim_{h \rightarrow 0^+} h^{a-1} \sin\left(\frac{1}{h}\right) = \lim_{h \rightarrow 0^+} \sin\left(\frac{1}{h}\right) = DNE$$

(Recall that the function  $\sin\left(\frac{1}{x}\right)$  oscillates faster and faster as  $x$  goes to 0. We first saw this behaviour in Example 1.3.5.)

- If  $a < 1$ , then  $a - 1 < 0$ , so  $\lim_{h \rightarrow 0^+} h^{a-1} = \infty$ . (Since we have a negative exponent, we are in effect *dividing by* a smaller and smaller positive number. For example, if  $a = \frac{1}{2}$ , then  $\lim_{h \rightarrow 0^+} h^{a-1} = \lim_{h \rightarrow 0^+} h^{-\frac{1}{2}} = \lim_{h \rightarrow 0^+} \frac{1}{\sqrt{h}} = \infty$ .) Since  $\sin\left(\frac{1}{x}\right)$  goes back and forth between one and negative one,

$$\lim_{h \rightarrow 0^+} h^{a-1} \sin\left(\frac{1}{x}\right) = DNE$$

since as  $h$  goes to 0, the function oscillates between positive and negative numbers of ever-increasing magnitude.

- If  $a > 1$ , then  $a - 1 > 0$ , so  $\lim_{h \rightarrow 0^+} h^{a-1} = 0$ . Although  $\sin\left(\frac{1}{x}\right)$  oscillates wildly near  $x = 0$ , it is bounded by  $-1$  and  $1$ . So,

$$(-1)h^{a-1} \leq h^{a-1} \sin\left(\frac{1}{h}\right) \leq h^{a-1}$$

Since both  $\lim_{h \rightarrow 0^+} (-1)h^{a-1} = 0$  and  $\lim_{h \rightarrow 0^+} h^{a-1} = 0$ , by the Squeeze Theorem,

$$\lim_{h \rightarrow 0^+} h^{a-1} \sin\left(\frac{1}{x}\right) = 0$$

as well.

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In the above cases, we learned

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} h^{a-1} \sin\left(\frac{1}{x}\right) = 0 \text{ when } a > 1, \text{ and}$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} h^{a-1} \sin\left(\frac{1}{x}\right) \neq 0 \text{ when } a \leq 1.$$

So,  $f$  is differentiable at  $x = 0$  if and only if  $a > 1$ .

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**Solutions to Exercises 2.3** — Jump to [TABLE OF CONTENTS](#)

**S-1:** (a) The slope of the secant line is  $\frac{h(24) - h(0)}{24 - 0} \frac{\text{m}}{\text{hr}}$ ; this is the change in height over the first day divided by the number of hours in the first day. So, it is the average rate of change of the height over the first day, measured in meters per hour.

(b) Consider (a). The secant line gives the *average* rate of change of the height of the dam; as we let the second point of the secant line get closer and closer to  $(0, h(0))$ , its slope approximates the instantaneous rate of change of the height of the water. So the slope of the tangent line is the instantaneous rate of change of the height of the water at the time  $t = 0$ , measured in  $\frac{\text{m}}{\text{hr}}$ .

**S-2:**  $p'(t) = \lim_{h \rightarrow 0} \frac{p(t+h) - p(t)}{h} \approx \frac{p(t+1) - p(t)}{1} = p(t+1) - p(t)$ , or the difference in profit caused by the sale of the  $(t+1)^{\text{st}}$  widget. So,  $p'(t)$  is the profit from the  $(t+1)^{\text{st}}$  widget. That is,  $p'(t)$  is the profit per additional widget sold, when  $t$  widgets are being sold. This is called the marginal profit per widget, when  $t$  widgets are being sold.

**S-3:** How quickly the temperature is changing per unit change of depth, measured in degrees per metre. In an ordinary body of water, the temperature near the surface ( $d = 0$ ) is pretty variable, depending on the sun, but deep down it is more stable (unless there are heat sources). So, one might reasonably expect that  $|T'(d)|$  is larger when  $d$  is near 0.

**S-4:**  $C'(w) = \lim_{h \rightarrow 0} \frac{C(w+h) - C(w)}{h} \approx \frac{C(w+1) - C(w)}{1} = C(w+1) - C(w)$ , which is the number of calories in  $C(w+1)$  grams minus the number of calories in  $C(w)$  grams. This is the number of calories per additional gram, when there are  $w$  grams.

**S-5:** The rate of change of velocity is acceleration. (If your velocity is increasing, you're accelerating; if your velocity is decreasing, you have negative acceleration.)

**S-6:** The rate of change in this case will be the relationship between the heat added and the temperature change.  $\lim_{h \rightarrow 0} \frac{T(j+h) - T(j)}{h} \approx \frac{T(j+1) - T(j)}{1} = T(j+1) - T(j)$ , or the change in temperature after the application of one joule. (This is closely related to heat capacity and to specific heat — there's a nice explanation of this on Wikipedia.)